

# Algebraic Geometry II

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In Algebraic Geometry I, we studied the classical approach of varieties in affine or projective space. While this is very geometric, there remain issues: Firstly, one needs the base field  $K$  to be algebraically closed: otherwise, Hilbert's Nullstellensatz fails, and the correspondence between geometric and algebraic objects breaks down (e.g.  $X^2 + 1 = 0$  has no solutions in  $\mathbb{R}$ ). In fact, we would like to not only look at non-algebraically closed fields, but more generally at algebraic sets over arbitrary rings, to study, for instance, algebraic equations over the integers using geometric methods.

Another issue of classical varieties comes from intersections: for example, take  $K = \mathbb{C}$  and look at  $V(y - x^2) \cap V(y) \subseteq \mathbb{A}^2$ . One can compute this intersection to be  $V(x^2, y) = V(x, y)$ , which corresponds to the single point  $(0, 0)$ . However, we would like to distinguish the algebraic sets  $V(x^2, y)$  and  $V(x, y)$ , since the former carries the extra information of "multiplicity". Algebraically, this corresponds to non-reduced  $K$ -algebras. If we had such a notion of multiplicity, we could prove more powerful theorems. For instance, Bézout's theorem simply reads  $|X \cap Y| = \deg X \deg Y$ , where the points in  $X \cap Y$  are counted with multiplicity.

To resolve these problems, and at the same time introduce a much more general and powerful framework, Grothendieck introduced the language of schemes. In the classical setting, note that the points of an affine variety  $X \subseteq \mathbb{A}^n$  correspond to the maximal ideals of  $A(X)$ . In the new setting, we will look at all prime ideals of  $A(X)$ .

For an arbitrary ring  $A$ , we can give  $\text{Spec } A$  a Zariski topology, and define regular functions on open subsets of  $\text{Spec } A$ . This is all similar to the classical ideas, but involves more technical objects. This will define the notion of an *affine scheme*.

## 1 Sheaves

**Definition 1.1.** Let  $X$  be a topological space. A *presheaf*  $\mathcal{F}$  of abelian groups on  $X$  consists of an abelian group  $\mathcal{F}(U)$  for each open subset  $U \subseteq X$  and a group homomorphism  $\rho_{U,V} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$  for any  $V \subseteq U$  open, such that  $\rho_{U,U} = \text{id}_{\mathcal{F}(U)}$  and  $\rho_{V,W} \circ \rho_{U,V} = \rho_{U,W}$  for any  $W \subseteq V \subseteq U$  open.

In other words, a presheaf is a contravariant functor from the category of open subsets of  $X$  to the category of abelian groups. The homomorphisms  $\rho_{U,V}$  are called *restriction maps*. An element  $s \in \mathcal{F}(U)$  is called a section of  $\mathcal{F}$  on  $U$ . If  $V \subseteq U$  is open, we will usually write  $s|_V := \rho_{U,V}(s)$ .

**Remark 1.2.** Analogously one defines presheaves of sets, groups, rings,  $R$ -modules etc.

**Example 1.3.** Let  $X$  be a topological space, and let  $\mathcal{F}(U) := \{f : U \rightarrow \mathbb{R} \mid f \text{ continuous}\}$ , with  $\rho_{U,V}$  the usual restriction of functions. Similarly, if  $X$  carries additional structure, one can look at conformal, differentiable or holomorphic functions on  $X$ .

Let  $X$  be a quasi-projective variety over a field  $K$ . Then the regular functions on  $X$  form a presheaf  $\mathcal{O}_X(U) = \{f : U \rightarrow K \mid f \text{ regular}\}$ .

In all examples, the presheaf  $\mathcal{F}$  satisfies the following two properties:

**Definition 1.4.** A presheaf  $\mathcal{F}$  on  $X$  is a *sheaf* if the following two conditions hold for any open  $U \subseteq X$  and every open cover  $U = \bigcup_{i \in I} U_i$ :

- (i) If  $s, t \in \mathcal{F}(U)$  with  $s|_{U_i} = t|_{U_i}$  for all  $i \in I$ , then  $s = t$ .
- (ii) If  $s_i \in \mathcal{F}(U_i)$  for all  $i \in I$  with  $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$  for all  $i, j \in I$ , then there exists  $s \in \mathcal{F}(U)$  with  $s|_{U_i} = s_i$  for all  $i \in I$ .

In fact, the section  $s$  in (ii) is unique by (i).

**Remark 1.5.** The examples in 1.3 are all sheaves. For an example of a presheaf that is not necessarily a sheaf, let  $1 \neq A$  be any abelian group,  $X$  a topological space, and  $\mathcal{F}(U) := A$  for all opens  $U \subseteq X$ . If  $X$  is not connected, this is not a sheaf. To rectify this, give  $A$  the discrete topology, and define the *constant sheaf*  $\underline{A}$  by

$$\underline{A}(U) := \text{Hom}(U, A) = \{f : U \rightarrow A \mid f \text{ locally constant}\}$$

**Definition 1.6.** Let  $\mathcal{F}$  be a presheaf on  $X$ , and let  $x \in X$ . Then the *stalk* of  $\mathcal{F}$  at  $x$  is defined as the direct limit

$$\mathcal{F}_x := \varinjlim_{x \in U} \mathcal{F}(U) = \{(U, s) \mid x \in U, s \in \mathcal{F}(U)\} / \sim,$$

where  $(U, s) \sim (V, t)$  if there exists an open  $x \in W \subseteq U \cap V$  with  $s|_W = t|_W$ .

**Remark 1.7.**  $\mathcal{F}_x$  is an abelian group with operation  $[(U, s)] + [(V, t)] = [(U \cap V, s|_{U \cap V} + t|_{U \cap V})]$ .

**Definition 1.8.** Let  $X$  be a topological space. A morphism  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  of (pre)sheaves on  $X$  is a family of morphisms  $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$  for all opens  $U \subseteq X$ , such that for all opens  $V \subseteq U$ , the following diagram commutes:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U) \\ \downarrow \rho_{UV} & & \downarrow \rho_{UV} \\ \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \end{array}$$

$\varphi$  is an isomorphism if there exists an inverse morphism  $\psi : \mathcal{G} \rightarrow \mathcal{F}$ .

**Remark 1.9.** In other words, a morphism of (pre)sheaves is a natural transformation of the two functors.

**Proposition 1.10.** Let  $x \in X$ . A morphism  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  of presheaves on  $X$  induces a morphism  $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$  of stalks, given explicitly by  $[(U, f)] \mapsto [(U, \varphi(U)(f))]$ .

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*Proof.* This follows directly from the universal property of direct limits.  $\square$

**Proposition 1.11.** Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of schemes. Then  $\varphi$  is an isomorphism if and only if  $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$  is an isomorphism for every  $x \in X$ .

Note that this fails for presheaves, and hence gives some motivation for the conditions in the definition of a sheaf.

*Proof.* If  $\varphi$  is an isomorphism, then so is  $\varphi_x$  (more precisely  $(\varphi_x)^{-1} = (\varphi_x)^{-1}$ ) by functoriality of direct limits.

Conversely, assume that  $\varphi_x$  is an isomorphism for every  $x \in X$ , and let  $U \subseteq X$  be open. Assume  $s \in \ker \varphi_U$ . For any  $x \in U$ , we have  $\varphi_x([(U, s)]) = [U, \varphi(U)(s)] = 0$ , so by injectivity of  $\varphi_x$  we get  $[(U, s)] = 0$  in  $\mathcal{F}_x$ . That is,  $s = 0$  on some neighbourhood  $x \in U_x \subseteq U$ . Then  $U = \bigcup_{x \in U} U_x$ , so by 1.4(1) we get  $s = 0 \in \mathcal{F}(U)$ . Hence  $\varphi_U$  is injective.

For surjectivity, let  $t \in \mathcal{G}(U)$ . For any  $x \in U$ ,  $\varphi_x$  is surjective, so let  $[(U_x, s_x)] \in \mathcal{F}_x$  be a preimage of  $[(U, t)] \in \mathcal{G}_x$ , wlog  $U_x \subseteq U$ . Then  $U = \bigcup_{x \in X} U_x$  and  $s_x \in \mathcal{F}(U_x)$ . To check the gluing condition, let  $x, y \in U$  and compute

$$\varphi_{U_x \cap U_y}(s_x|_{U_x \cap U_y}) \stackrel{1.8}{=} (\varphi_{U_x}(s_x))|_{U_x \cap U_y} = t|_{U_x}|_{U_x \cap U_y} \stackrel{1.1}{=} t|_{U_x \cap U_y} = \varphi_{U_x \cap U_y}(s_y|_{U_x \cap U_y}),$$

thus by injectivity  $s_x|_{U_x \cap U_y} = s_y|_{U_x \cap U_y}$ . Hence by 1.4(2) we obtain an element  $s \in \mathcal{F}(U)$  s.t.  $s|_{U_x} = s_x$  for every  $x \in X$ . It remains to check that  $\varphi_U(s) = t$ . But this holds by 1.4(1) applied to  $\mathcal{G}$ , since by construction  $\varphi_U(s)|_{U_x} = \varphi_{U_x}(s|_{U_x}) = \varphi_{U_x}(s_x) = t|_{U_x}$ . Therefore,  $\varphi_U$  is surjective, hence an isomorphism.  $\square$

**Definition 1.12** (Naive<sup>1</sup>). Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of presheaves. We say that  $\varphi$  is injective/surjective if for every  $x \in X$ ,  $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$  is injective/surjective. In particular, if  $\mathcal{F}, \mathcal{G}$  are sheaves, then by proposition 1.11,  $\varphi$  is an isomorphism if and only if it is injective and surjective.

**Definition 1.13.** Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of presheaves on  $X$ . We define the following presheaves:

$$\begin{aligned} \ker(\varphi)(U) &:= \ker(\varphi_U), \\ \operatorname{im}(\varphi)^{\operatorname{pre}}(U) &:= \operatorname{im}(\varphi_U) \\ \operatorname{coker}(\varphi)^{\operatorname{pre}}(U) &:= \operatorname{coker}(\varphi_U) \end{aligned}$$

**Remark 1.14.** In each case, one easily checks that these assignments define presheaves. If  $\mathcal{F}, \mathcal{G}$  are sheaves, then  $\ker \varphi$  is a sheaf as well (Exercise), but in general  $\operatorname{im} \varphi$  and  $\operatorname{coker} \varphi$  are not sheaves.

**Proposition 1.15** (Sheafification). *Let  $\mathcal{F}$  be a presheaf on  $X$ . Then there is a unique (up to unique isomorphism) sheaf  $\mathcal{F}^+$ , called sheafification of  $\mathcal{F}$ , and a morphism  $\theta : \mathcal{F} \rightarrow \mathcal{F}^+$  with the following universal property: For any sheaf  $\mathcal{G}$  and morphism  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  of presheaves, there exists a unique morphism  $\tilde{\varphi} : \mathcal{F}^+ \rightarrow \mathcal{G}$  making the following diagram commute:*

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\varphi} & \mathcal{G} \\ & \searrow \theta & \nearrow \tilde{\varphi} \\ & \mathcal{F}^+ & \end{array}$$

Moreover,  $\theta$  induces isomorphisms  $\theta_x : \mathcal{F}_x \rightarrow \mathcal{F}_x^+$  for every  $x \in X$ . If  $\mathcal{F}$  is already a sheaf, then  $\theta$  is an isomorphism.

*Proof.* We construct  $\mathcal{F}^+$  as follows:

$$\mathcal{F}^+(U) := \left\{ s : U \rightarrow \prod_{x \in U} \mathcal{F}_x \mid s(x) \in \mathcal{F}_x \text{ for all } x \in U, \text{ and } (*) \right\},$$

where  $(*)$  is the following extra condition: For every  $x \in U$  there exists an open neighbourhood  $x \in V \subseteq U$  and  $t \in \mathcal{F}(V)$  with  $s(y) = t_y := [(V, t)]$  for all  $y \in V$ . Further, set  $\theta_U(t)(x) = t_x$  for all  $U \subseteq X$  open,  $t \in \mathcal{F}(U)$  and  $x \in U$ . One checks that this  $\mathcal{F}^+$ ,  $\theta$  have the required properties:

It is clear that  $\mathcal{F}^+$  is a sheaf, since it is constructed via functions with local properties, and that  $\theta$  is a morphism of presheaves, since  $(t|_W)_x = t_x$  for  $x \in W \subseteq V$ ,  $t \in \mathcal{F}(V)$ . Let  $x \in X$  and  $[(V, t)] \in \ker \theta_x$ . Up to shrinking  $V$ , we have  $\theta_V(t) = 0$ . In particular,  $[(V, t)] = \theta_V(t)(x) = 0$ . Let  $[(V, s)] \in \mathcal{F}_x^+$ . By  $(*)$  wlog there is  $t \in \mathcal{F}(V)$  s.t.  $s(y) = t_y$  for all  $y \in V$ . Hence  $\theta_V(t) = [(V, s)]$ . If  $\mathcal{F}$  is a sheaf, then  $\theta$  is thus an isomorphism by proposition 1.11.

Finally, we check the universal property: Let  $\mathcal{G}$  be a sheaf, and  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of presheaves. By the previous paragraph, it suffices to give a morphism  $\tilde{\varphi} : \mathcal{F}^+ \rightarrow \mathcal{G}^+ \cong \mathcal{G}$ . Let  $U \subseteq X$  be open and let  $\tilde{\varphi}(s)(y) = \varphi_y(s(y))$ . It is easy to check that this is a well-defined morphism of sheaves, makes the diagram commute, and is the only morphism with this property. The details are left as an exercise.  $\square$

<sup>1</sup>will be corrected in 1.21

**Example 1.16.** Let  $X$  be a topological space such that for every  $x \in X$  there is a neighbourhood  $x \in U_x \subsetneq X$ , and let  $0 \neq A$  an abelian group.

- (i) Let  $\mathcal{F}(X) = A$  and  $\mathcal{F}(U) = 0$  otherwise. This is clearly not a sheaf. By the assumption on  $X$  we have  $\mathcal{F}_x = 0$  for all  $x \in X$ , so  $\mathcal{F}^+ = 0$ .
- (ii) Let  $\mathcal{F}$  be the constant presheaf with value  $A$  as in remark 1.5. From the universal property one sees that  $\mathcal{F}^+ = \underline{A}$  is the constant sheaf.

**Definition 1.17.** A sub-(pre)sheaf of a (pre)sheaf  $\mathcal{F}$  on  $X$  is a (pre)sheaf  $\mathcal{F}'$  s.t. for every open  $U \subseteq X$  one has  $\mathcal{F}'(U) \subseteq \mathcal{F}(U)$ , and for every  $V \subseteq U$  open,  $\rho'_{UV} = \rho_{UV}|_{\mathcal{F}'(U)}$ . In other words, the inclusions  $\mathcal{F}'(U) \hookrightarrow \mathcal{F}(U)$  have to form a morphism of (pre)sheaves.

If  $\mathcal{F}' \subseteq \mathcal{F}$  is a sub-presheaf, then for every  $x \in X$  we have  $\mathcal{F}'_x \subseteq \mathcal{F}_x$ .

**Proposition 1.18.** Let  $\mathcal{F}$  be a sheaf and  $\mathcal{F}'$  be a subsheaf. Then  $\mathcal{F}' = \mathcal{F}$  if and only if  $\mathcal{F}'_x = \mathcal{F}_x$  for all  $x \in X$ .

*Proof.* Follows directly from proposition 1.11.  $\square$

**Example 1.19.** If  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  is a morphism of presheaves, then  $\ker \varphi$  and  $\operatorname{im} \varphi^{\text{pre}}$  are sub-presheaves of  $\mathcal{F}$  resp.  $\mathcal{G}$ .

**Exercise 1.20.** Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of presheaves. Show that  $(\ker \varphi)_x = \ker(\varphi_x)$  and  $(\operatorname{im} \varphi^{\text{pre}})_x = \operatorname{im}(\varphi_x)$ .

**Definition 1.21.** Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of presheaves. We say that  $\varphi$  is injective if and only if  $\ker \varphi = 0$ . For presheaves, we say  $\varphi$  is surjective if  $\operatorname{coker} \varphi^{\text{pre}} = 0$ <sup>2</sup>. If  $\mathcal{F}, \mathcal{G}$  are sheaves, define the image and cokernel sheaves via  $\operatorname{im} \varphi := (\operatorname{im} \varphi^{\text{pre}})^+$  and  $\operatorname{coker} \varphi := (\operatorname{coker} \varphi^{\text{pre}})^+$ . Then  $\operatorname{im} \varphi$  can be identified with a subsheaf of  $\mathcal{G}$  (see next proposition). We say that  $\varphi$  is surjective as a morphism of sheaves if  $\operatorname{im} \varphi = \mathcal{G}$ , equivalently  $\operatorname{coker} \varphi = 0$ .

**Proposition 1.22.** Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of sheaves. Then  $\operatorname{im} \varphi$  can be seen as a subsheaf of  $\mathcal{G}$ . Moreover,  $\operatorname{im} \varphi = \mathcal{G} \iff \varphi_x$  is surjective for all  $x \in X$ .

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*Proof.* We have  $\operatorname{im} \varphi \subseteq \mathcal{G}^+ \cong \mathcal{G}$ . The second part follows directly from proposition 1.11.  $\square$

**Remark 1.23.** Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of (pre)sheaves. Then  $\varphi$  is injective if and only if  $\varphi_U$  is injective for all  $U \subseteq X$  open, that is  $\ker \varphi = 0$ . If  $\varphi$  is injective, then also  $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$  is injective, but the converse is only true for sheaves.

$\varphi$  is a surjective morphism of presheaves if  $\varphi_U$  is surjective for all  $U \subseteq X$  open, that is  $\operatorname{im} \varphi^{\text{pre}} = \mathcal{G}$ . Again, if  $\varphi$  is surjective, then so are all  $\varphi_x$ , but the converse is not true in general (even for sheaves).

If  $\mathcal{F}, \mathcal{G}$  are sheaves,  $\varphi$  is a surjective morphism of sheaves if and only if  $\operatorname{im} \varphi = \mathcal{G}$ , if and only if  $\varphi_x$  is surjective for all  $x \in X$ .

**Example 1.24.** (i) Let  $\mathcal{F}, \mathcal{G}$  be (pre)sheaves on  $X$ . Then  $(\mathcal{F} \oplus \mathcal{G})(U) := \mathcal{F}(U) \oplus \mathcal{G}(U)$  is again a (pre)sheaf.

(ii) Let  $A$  be an abelian group,  $X$  a topological space and  $x \in X$ . The sheaf

$${}^x\mathcal{F}(U) := \begin{cases} A, & \text{if } x \in U, \\ 0 & \text{otherwise} \end{cases}$$

is called the *skyscraper sheaf*. One calculates  ${}^x\mathcal{F}_y = A$  if  $y \in \overline{\{x\}}$ , and  ${}^x\mathcal{F}_y = 0$  otherwise. In particular, if  $X$  is  $T_1$  (that is, points are closed), then  ${}^x\mathcal{F}_y \neq 0$  only for  $x = y$ .

<sup>2</sup>Note that this is not equivalent to  $\varphi_X$  being surjective for all  $x \in X$ .

- (iii) Let  $X$  be a connected  $T_1$ -space,  $x \neq z \in X$ . Let  $0 \neq A$  be an abelian group, and consider the natural map  $\varphi : \underline{A} \rightarrow {}^x\mathcal{F} \oplus {}^z\mathcal{F}$ . One sees that  $\varphi$  is surjective as a morphism of sheaves, but not as a morphism of presheaves.

**Definition 1.25.** Let  $f : X \rightarrow Y$  be a continuous map of topological spaces.

- (i) Let  $\mathcal{F}$  be a sheaf on  $X$ . The *direct image* of  $\mathcal{F}$  is the sheaf  $f_*\mathcal{F}$  on  $Y$  defined by

$$f_*\mathcal{F}(U) := \mathcal{F}(f^{-1}(U)).$$

- (ii) Let  $\mathcal{G}$  be a sheaf on  $Y$ . The *inverse image*  $f^{-1}\mathcal{G}$  of  $\mathcal{G}$  is defined as the sheafification of the presheaf

$$U \mapsto \varinjlim_{V \supseteq f(U)} \mathcal{G}(V).$$

## 2 Affine Schemes

In Algebraic Geometry I, we showed an equivalence of categories between affine algebraic sets over  $K$ , and finitely generated reduced  $K$ -algebras, for  $K$  algebraically closed. We want to generalize this to allow any ring  $A$  on the algebraic side. A natural generalization of the classical setting is to consider  $\text{Spec } A$ , and to endow this set with topological and algebraic information.

Let  $A$  be a ring (always commutative with 1).

**Definition 2.1.**  $\text{Spec } A := \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ prime ideal}\}$ . If  $I \subseteq A$  is any ideal, let  $V(I) := \{\mathfrak{p} \in \text{Spec } A \mid I \subseteq \mathfrak{p}\}$ . (Note that there is a canonical bijection  $V(I) \cong \text{Spec } A/I$ .) For  $S \subseteq A$  any subset, we write  $V(S) := V(\langle S \rangle)$ .

**Lemma 2.2.** Let  $I, J \subseteq A$  be ideals.

- (i)  $\sqrt{I} := \{f \in A \mid f^n \in I \text{ for some } n \geq 1\} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p}$ .
- (ii)  $I \subseteq \sqrt{I}$  and  $V(I) = V(\sqrt{I})$ .
- (iii) If  $I \subseteq J$ , then  $V(J) \subseteq V(I)$ .
- (iv) If  $V(J) \subseteq V(I)$  if and only if  $\sqrt{I} \subseteq \sqrt{J}$ .

*Proof.* (i) was proven in Commutative Algebra. (ii), (iii), and one direction of (iv) are clear. So assume  $V(J) \subseteq V(I)$ , then  $\sqrt{I} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p} \subseteq \bigcap_{\mathfrak{p} \in V(J)} \mathfrak{p} = \sqrt{J}$  by (i).  $\square$

**Lemma 2.3.** Let  $I, J, \{I_\alpha\}_{\alpha \in I} \subseteq A$  be ideals.

- (i)  $V(IJ) = V(I \cap J) = V(I) \cup V(J)$ .
- (ii)  $V(\sum_\alpha I_\alpha) = \bigcap_\alpha V(I_\alpha)$ .

*Proof.* (i) For a prime ideal  $\mathfrak{p} \subseteq A$ , we have  $IJ \subseteq \mathfrak{p}$  if and only if  $I \subseteq \mathfrak{p}$  or  $J \subseteq \mathfrak{p}$ , if and only if  $I \cap J \subseteq \mathfrak{p}$ .

(ii) Exercise.  $\square$

**Definition 2.4.** By lemma 2.3 (and the observations  $V(A) = \emptyset$ ,  $V(0) = \text{Spec } A$ ), the collection  $\{V(I) \mid I \subseteq A \text{ ideal}\}$  forms the set of closed sets of a topology on  $\text{Spec } A$ , the so-called *Zariski topology*.

**Remark 2.5.** Not every point of  $\text{Spec } A$  is closed. More precisely, let  $x \in \text{Spec } A$ . Then  $\overline{\{x\}} = V(x)$ , so  $\{x\}$  is closed if and only if  $x$  is a maximal ideal.

**Example 2.6.** (i) If  $A$  is an integral domain, then  $0 \in \text{Spec } A$  satisfies  $\overline{\{0\}} = \text{Spec } A$ . In particular, every open in  $\text{Spec } A$  contains  $\{0\}$ . Such a point is called a *generic point* of  $\text{Spec } A$ .

(ii) Let  $\mathbb{A}^1 := \text{Spec } K[x]$ . Then  $\mathbb{A}^1$  contains the generic point  $(0)$ , plus closed points corresponding to normalised irreducible polynomials in  $K[x]$ . If  $K$  is algebraically closed, every irreducible polynomial is linear, so  $\mathbb{A}^1$  looks very similar to affine space in the classical setting, the only difference being the additional generic point.

(iii)  $\text{Spec } \mathbb{Z} = \{(0)\} \cup \{(p) \mid p \text{ prime}\}$

(iv) Let  $\mathbb{A}^2 := \text{Spec } K[x, y]$  with  $K$  algebraically closed. It contains the generic point  $(0)$ , closed points  $(x - a, y - b)$  as in the classical settings, but also (principal) prime ideals corresponding to curves. Let  $(F)$  be such an ideal, then

$$\overline{\{(F)\}} = \{(F)\} \cup \{(x - a, y - b) \mid F(a, b) = 0\}.$$

**Lemma 2.7.** (i) If  $A$  is an integral domain, then  $\text{Spec } A$  is irreducible. (The converse fails e.g. for  $K[x]/(x^2)$ ). If  $A$  is reduced and  $\text{Spec } A$  is irreducible, then  $A$  is an integral domain.

- (ii)  $\text{Spec } A$  is not connected if and only if there exists  $0, 1 \neq e \in A$  with  $e^2 = e$ , if and only if  $A \cong A_1 \times A_2$  with  $A_1, A_2 \neq 0$ .

*Proof.* (i) Assume that  $\text{Spec } A$  is irreducible and  $A$  is reduced but not an integral domain. Let  $a * b = 0$  with  $0 \neq a, b \in A$ . Then  $\text{Spec } A = V(0) = V(a) \cup V(b)$ . If  $V(a) = \text{Spec } A$ , then  $a \in \sqrt{0} = 0$  by assumption, contradiction. Hence  $\text{Spec } A$  is reducible. The rest is left as an exercise.  $\square$

**Definition 2.8.** Let  $f \in A$ . Then  $D(f) := \text{Spec } A \setminus V(f) = \{\mathfrak{p} \in \text{Spec } A \mid f \notin \mathfrak{p}\}$  is called a *distinguished* or *standard open* subset of  $\text{Spec } A$ .

**Remark 2.9.** One has  $D(f) \cap D(g) = D(fg)$ . The set  $\{D(f) \mid f \in A\}$  forms a basis of the Zariski topology on  $\text{Spec } A$ : Let  $U = \text{Spec } A \setminus V(I)$  be open and  $\mathfrak{p} \in U$ . Since  $\mathfrak{p} \notin V(I)$ ,  $I \not\subseteq \mathfrak{p}$ . Hence choose  $f \in I \setminus \mathfrak{p}$ , which satisfies  $\mathfrak{p} \in D(f) \subseteq U$ .

We want to construct a sheaf of "regular functions" on  $\text{Spec } A$ . Recall from the classical setting, that if  $X \subseteq \mathbb{A}^n$  was an affine algebraic set and  $U \subseteq X$  open, we defined  $\mathcal{O}_X(U)$  as the set of all functions  $f : U \rightarrow K$  that locally are rational functions, and we proved  $\mathcal{O}_X(D(f)) \cong A(X)_f$  and  $\mathcal{O}_{X,x} \cong \varinjlim_{D(f) \ni x} A(X)_f \cong A(X)_{\mathfrak{m}_x}$ . This motivates us to define a sheaf  $\mathcal{O}_X$  of rings on  $\text{Spec } A$  s.t.  $\mathcal{O}_X(D(f)) \cong A_f$ . Note that if  $D(g) \subseteq D(f)$ , then  $g^m = fa$  for some  $a \in A, m \geq 1$  by lemma 2.2, so there is a natural map  $A_f \rightarrow A_g$ . Since we know what the stalks are supposed to look like, we can proceed as in the proof of sheafification 1.15.

**Definition 2.10.** Let  $X = \text{Spec } A$ . The structure sheaf  $\mathcal{O}_X$  is defined as

$$\mathcal{O}_X(U) := \left\{ s : U \rightarrow \prod_{\mathfrak{p} \in U} A_{\mathfrak{p}} \mid s(\mathfrak{p}) \in A_{\mathfrak{p}} \text{ and } (*) \right\},$$

where  $(*)$  is the extra condition that for every  $\mathfrak{p} \in U$  there exists an open neighbourhood  $V \subseteq U$  and  $a, f \in A$  such that  $s(\mathfrak{q}) = \frac{a}{f}$  in  $A_{\mathfrak{q}}$  for every  $\mathfrak{q} \in V$ . By construction it is clear that  $\mathcal{O}_X$  is a sheaf of rings on  $\text{Spec } A$ .

**Proposition 2.11.** For  $\mathfrak{p} \in \text{Spec } A$  we have  $\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}$ , and for  $f \in A$ ,  $\mathcal{O}_X(D(f)) \cong A_f$ . In particular,  $\mathcal{O}_X(X) \cong A$ .

*Proof.* Let  $\mathfrak{p} \in \text{Spec } A$  and set  $\varphi : \mathcal{O}_{X,\mathfrak{p}} \rightarrow A_{\mathfrak{p}}, [(U, f)] \mapsto f(\mathfrak{p})$ . This is a well-defined homomorphism of rings. Let  $\frac{a}{s} \in A_{\mathfrak{p}}$  with  $a, s \in A, s \notin \mathfrak{p}$ . That is,  $\mathfrak{p} \in D(s)$ , so  $[(D(s), \frac{a}{s})]$  is a well-defined preimage of  $\frac{a}{s}$ . Conversely, let  $[(U, f)] \in \ker \varphi$ . By shrinking  $U$ , we may assume  $f(\mathfrak{q}) = \frac{a}{s} \in A_{\mathfrak{q}}, a, s \in A$ , for all  $\mathfrak{q} \in U$ . Then  $0 = \varphi([(U, f)]) = \frac{a}{s} \in A_{\mathfrak{p}}$ , hence  $s'a = 0$  for some  $s' \notin \mathfrak{p}$ , i.e.  $\mathfrak{p} \in D(s')$ . By shrinking  $U$  further, if necessary, we may assume  $U \subseteq D(s')$ . Then  $f = \frac{a}{s} = \frac{s'a}{s's} = 0$ . Therefore,  $\varphi$  is an isomorphism  $\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}$ .

Now let  $f \in A$ . Define  $\psi : A_f \rightarrow \mathcal{O}_X(D(f)), \alpha \mapsto (t_{\alpha} : \mathfrak{q} \mapsto \alpha)$ . This is clearly a well-defined homomorphism of rings. Let  $\alpha = \frac{a}{f^n} \in \ker \psi$ . That is,  $\frac{a}{f^n} = 0$  in  $A_{\mathfrak{q}}$  for every  $\mathfrak{q} \in D(f)$ . This means that for every  $\mathfrak{q} \in D(f)$  there exists  $s_{\mathfrak{q}} \notin \mathfrak{q}$  with  $s_{\mathfrak{q}}a = 0$ . Consider the ideal  $I = \{b \in A \mid ba = 0\}$  (called the annihilator of  $a$ ). Then for every  $\mathfrak{q} \in D(f)$ ,  $s_{\mathfrak{q}} \in I \setminus \mathfrak{q}$ , i.e.  $I \not\subseteq \mathfrak{q}$ . Hence  $V(I) \subseteq \text{Spec } A \setminus D(f) = V(f)$ . By lemma 2.2 it follows that  $f^m \in I$  for some  $m > 0$ . Then  $\alpha = \frac{a}{f^n} = \frac{af^m}{f^{n+m}} = 0$ .

For surjectivity, let  $t \in \mathcal{O}_X(D(f))$ . By definition, there is an open cover  $D(f) = \bigcup_i V_i$  s.t.  $t|_{V_i} = \frac{a_i}{s_i}$  for some  $a_i, s_i \in A$  with  $V_i \subseteq D(s_i)$ . By remark 2.9 wlog  $V_i = D(h_i)$ . Now  $V(h_i) \supseteq V(s_i)$ , so again by lemma 2.2,  $h_i \in \sqrt{(s_i)}$ , i.e.  $h_i^{n_i} = s_i c_i$  for some  $n_i > 0, c_i \in A$ . It follows that  $t|_{D(h_i)} = \frac{a_i}{s_i} = \frac{a_i c_i}{h_i^{n_i}}$ . Replacing  $h_i^{n_i}$  by  $h_i$  (since  $D(h_i) = D(h_i^{n_i})$ ) and  $a_i c_i$  by  $a_i$ , we simply write  $t|_{D(h_i)} = \frac{a_i}{h_i}$ .



Observe that there exists a finite subcover  $D(f) = \bigcup_{i=1}^k D(h_i)$  of  $\bigcup_i D(h_i)$ . Indeed,  $D(f) \subseteq \bigcup_i D(h_i)$  implies  $V(f) \supseteq \bigcap_i V(h_i) = V(\sum_i (h_i))$  by lemma 2.3. By lemma 2.2,  $f^n \in \sum_i (h_i)$  for some  $n > 0$ . But this means  $f^n$  is a *finite* linear combination of the  $h_i$ 's. Taking only those  $h_i$  which appear in this sum and retracing the argument backwards yields the claim.

For  $i \neq j$ , we have  $\frac{a_i h_j}{h_i h_j} = t|_{D(h_i) \cap D(h_i h_j)} = t|_{D(h_j) \cap D(h_i h_j)} = \frac{a_j h_i}{h_j h_i}$  as functions on  $D(h_i h_j)$ . By the injectivity of  $\psi$  (for  $D(h_i h_j)$ ) proved above, even  $\frac{a_i h_j}{h_i h_j} = \frac{a_j h_i}{h_j h_i}$  as elements of  $A_{h_i h_j}$ , i.e.  $(h_i h_j)^n (a_i h_j - a_j h_i) = 0$  in  $A$ , for some  $n$ . Since there are only finitely many sets in the open cover, we can choose  $n$  big enough such that this equality holds for all  $i, j$ .

Replacing  $h_i$  by  $h_i^{n+1}$  and  $a_i$  by  $h_i^n a_i$ , we still have  $t|_{D(h_i)} = \frac{a_i}{h_i}$ , and now  $h_j a_i = h_i a_j$  for all  $i, j$ . As before,  $f \in \sqrt{(h_1, \dots, h_k)}$ , so we can write  $f^m = \sum_i b_i h_i$  for some  $m > 0$  and  $b_i \in A$ . Let  $a := \sum_i b_i a_i$ . We claim that  $\psi(\frac{a}{f^m}) = t$ . Indeed, for every  $j$  we have  $f^m a_j = \sum_i b_i h_i a_j = \sum_i b_i a_i h_j = a h_j$ , i.e.  $\frac{a}{f^m} = \frac{a_j}{h_j} \in A_q$  for every  $q \in D(h_j)$ .  $\square$

Lecture 7  
May 4, 2026

Having now associated to a ring  $A$  the pair  $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ , we next want to define morphisms between these objects. We let us motivate by the following

**Remark 2.12.** Recall from classical algebraic geometry that if  $X, Y$  are quasi-projective algebraic sets, a map  $\varphi : X \rightarrow Y$  was defined to be a morphism if  $\varphi$  is continuous, and for every open  $U \subseteq Y$  and  $f \in \mathcal{O}_Y(U)$ , its preimage  $\varphi^* f = f \circ \varphi$  is regular, i.e. lies in  $\mathcal{O}_X(\varphi^{-1}(U))$ . This yielded compatible homomorphisms of  $K$ -algebras  $\mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(\varphi^{-1}(U))$ , in other words a morphism of sheaves  $\mathcal{O}_Y \rightarrow \varphi_* \mathcal{O}_X$ .

- Definition 2.13.** (i) A *ringed space* is a pair  $(X, \mathcal{O}_X)$ , where  $X$  is a topological space and  $\mathcal{O}_X$  a sheaf of rings on  $X$ .
- (ii) A morphism of ringed spaces  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  is a pair  $(f, f^\#)$ , where  $f : X \rightarrow Y$  is a continuous map, and  $f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$  is a morphism of sheaves.
- (iii) A *locally ringed space* is a ringed space  $(X, \mathcal{O}_X)$  such that all stalks  $\mathcal{O}_{X,x}$ ,  $x \in X$  are local rings.
- (iv) A morphism of locally ringed spaces  $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  is a morphism of ringed spaces such that for every  $x \in X$ , the map  $f_x^\# : \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$  is a local homomorphism of rings (where a morphism of rings  $\varphi : (A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$  of local rings is called *local* if  $\varphi^{-1}(\mathfrak{n}) = \mathfrak{m}$ ). The map  $f_x^\#$  is defined as the compositum

$$\mathcal{O}_{Y,f(x)} \xrightarrow{f_{f(x)}^\#} (f_* \mathcal{O}_X)_{f(x)} = \varinjlim_{f^{-1}(U) \ni x} \mathcal{O}_X(f^{-1}(U)) \xrightarrow{\text{incl}_*} \varinjlim_{U \ni x} \mathcal{O}_X(U) = \mathcal{O}_{X,x},$$

where the second map is induced by inclusion of directed sets.

**Remark 2.14.** Let  $A$  be a ring. The pair  $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$  is a locally ringed space by proposition 2.11.

- Definition 2.15.** (i) Let  $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  and  $(g, g^\#) : (Y, \mathcal{O}_Y) \rightarrow (Z, \mathcal{O}_Z)$  be morphisms of (locally) ringed spaces. Then their composition is defined as  $(g, g^\#) \circ (f, f^\#) := (g \circ f, (g \circ f)^\#)$ , with  $(g \circ f)^\# := g_* f^\# \circ g^\# : \mathcal{O}_Z \rightarrow (g \circ f)_* \mathcal{O}_X$ .
- (ii) An isomorphism of (locally) ringed spaces is a morphism with a two-sided inverse morphism.

A morphism  $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  is an isomorphism if and only if  $f : X \rightarrow Y$  is a homeomorphism and  $f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$  is an isomorphism of sheaves. (Exercise)

**Lemma 2.16.** Let  $\varphi : A \rightarrow B$  be a morphism of rings. Then  $f : \text{Spec } B \rightarrow \text{Spec } A$ ,  $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$  is continuous.

*Proof.*  $f$  is well-defined since the preimage of a prime ideal is again prime. Let  $V(I) \subseteq \operatorname{Spec} A$ . Then  $f^{-1}(V(I)) = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi^{-1}(\mathfrak{p}) \in V(I)\} = V(\varphi(I))$  is closed.  $\square$

**Remark 2.17.** Let  $\varphi : A \rightarrow B$  be a ring morphism and  $X = \operatorname{Spec} B, Y = \operatorname{Spec} A$ . Then  $\varphi$  induces a morphism of locally ringed spaces  $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ , where  $f$  is the map of lemma 2.16, and  $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$  is defined on  $U \subseteq Y$  as

$$f_U^\# : \mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(f^{-1}(U)), \quad t \mapsto (\mathfrak{p} \mapsto \varphi_{\mathfrak{p}}(t(f(\mathfrak{p})))),$$

where  $\varphi_{\mathfrak{p}} : A_{f(\mathfrak{p})} \rightarrow B_{\mathfrak{p}}$  is the natural morphism induced by the universal property of localisation. Note that if  $t|_V = \frac{a}{s}$ , then  $f^\#(t)|_{f^{-1}(V)} = \frac{\varphi(a)}{\varphi(s)}$ , so  $f^\#$  is indeed a well-defined morphism of sheaves.

We check that  $(f, f^\#)$  is a morphism of locally ringed spaces, i.e. that  $f_{\mathfrak{p}}^\# : \mathcal{O}_{Y, f(\mathfrak{p})} \rightarrow (f_*\mathcal{O}_X)_{f(\mathfrak{p})} \rightarrow \mathcal{O}_{X, \mathfrak{p}}$  is a local morphism of rings. By definition, this map is given by  $[(U, t)] \mapsto [(f^{-1}(U), f^\#(t))]$ , and hence by construction the diagram

$$\begin{array}{ccc} \mathcal{O}_{Y, f(\mathfrak{p})} & \xrightarrow{f_{\mathfrak{p}}^\#} & \mathcal{O}_{X, \mathfrak{p}} \\ \downarrow \cong & & \downarrow \cong \\ A_{f(\mathfrak{p})} & \xrightarrow{\varphi_{\mathfrak{p}}} & B_{\mathfrak{p}} \end{array}$$

commutes, where the vertical isomorphisms come from proposition 2.11. Since  $\varphi_{\mathfrak{p}}$  is local, hence so is  $f_{\mathfrak{p}}^\#$ .

As was the case in classical algebraic geometry, we have

**Proposition 2.18.** *Let  $A, B$  be rings, and  $X = \operatorname{Spec} B, Y = \operatorname{Spec} A$ . Then the construction above yields a bijection*

$$\operatorname{Hom}_{\mathbf{Ring}}(A, B) \xrightarrow{\cong} \operatorname{Hom}_{\mathbf{LRS}}((X, \mathcal{O}_X), (Y, \mathcal{O}_Y)).$$

(In other words, the functor  $\operatorname{Spec}$  is fully faithful.)

*Proof.* Let  $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  be a morphism of locally ringed spaces. Then  $f_Y^\# : \mathcal{O}_Y(Y) \cong A \rightarrow B \cong f_*\mathcal{O}_X(Y)$ , we claim that this construction is inverse to the one constructed in remark 2.17. Given  $\varphi : A \rightarrow B$ , let  $(f, f^\#)$  be the induced morphism of locally ringed spaces. Then  $f_Y^\# = \varphi$  (under the canonical identifications) by construction.

Conversely, let  $(f, f^\#)$  be a morphism of locally ringed spaces, and let  $(g, g^\#)$  be the morphism constructed from  $\varphi := f_Y^\#$  as above. Let  $\mathfrak{p} \in X$ . We have a commutative diagram

$$\begin{array}{ccc} B \cong \mathcal{O}_X(X) & \xleftarrow{f_Y^\#} & \mathcal{O}_Y(Y) \cong A \\ \downarrow & & \downarrow \\ B_{\mathfrak{p}} \cong \mathcal{O}_{X, \mathfrak{p}} & \xleftarrow{f_{\mathfrak{p}}^\#} & \mathcal{O}_{Y, f(\mathfrak{p})} \cong A_{f(\mathfrak{p})} \end{array}$$

Since  $f_{\mathfrak{p}}^\#$  is a local morphism, the preimage of  $\mathfrak{p}B_{\mathfrak{p}}$  is the image of  $f(\mathfrak{p})$  in  $A_{f(\mathfrak{p})}$ . By commutativity of the diagram, this equals  $\varphi^{-1}(\mathfrak{p})$ , i.e.  $g = \varphi^{-1} = f$ .

By the universal property of localizations, there is a unique map  $A_{f(\mathfrak{p})} \rightarrow B_{\mathfrak{p}}$  making the above diagram commute. But both  $f_{\mathfrak{p}}^\#$  and  $\varphi_{\mathfrak{p}}$  do, so they are equal. By the construction in remark 2.17, this immediately implies  $f^\# = g^\#$ .  $\square$

**Definition 2.19.** An *affine scheme* is a locally ringed space isomorphic to  $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$  for some ring  $A$ .

**Remark 2.20.** Immediately from the definition and proposition 2.18 it follows that the functor  $\text{Spec} : \text{Ring} \rightarrow \text{AffSch}$ ,  $A \mapsto \text{Spec } A$ ,  $\varphi \mapsto (f, f^\#)$  as in remark 2.17 is an (anti-)equivalence of categories.

**Definition 2.21.** A *scheme* is a locally ringed space  $(X, \mathcal{O}_X)$  such that for every  $x \in X$  there exists a neighbourhood  $x \in U \subseteq X$  such that  $(U, \mathcal{O}_X|_U)$  is an affine scheme.

By abuse of notation, we will often write  $X$  instead of  $(X, \mathcal{O}_X)$  and  $f$  instead of  $(f, f^\#)$ . Given a scheme  $X$ , we will write  $\text{Sp}(X)$  for the underlying topological space.

Lecture 9  
May 11, 2026

**Lemma 2.22.** Let  $A$  be a ring and  $X = \text{Spec } A$ . Let  $g \in A$  and consider  $D(g) \subseteq \text{Spec } A$ . Then  $(D(g), \mathcal{O}_X|_{D(g)}) \cong \text{Spec } A_g$  is again an affine scheme.

*Proof.* Exercise. □

**Proposition 2.23.** Let  $X$  be a scheme and  $U \subseteq X$  be open. Then  $(U, \mathcal{O}_X|_U)$  is again a scheme.

*Proof.*  $(U, \mathcal{O}_X|_U)$  is clearly a locally ringed space. We only need to check that locally it is an affine scheme, so wlog  $X = \text{Spec } A$  is affine. Let  $U = \bigcup_i D(g_i)$  be an open cover by standard opens, then each  $(D(g_i), \mathcal{O}_X|_{D(g_i)}) = (D(g_i), \mathcal{O}_X|_U|_{D(g_i)})$  is an affine scheme by the previous lemma, so the result follows immediately. □

**Definition 2.24.** (i) Let  $X$  be a scheme and  $U \subseteq X$  be open. Then  $(U, \mathcal{O}_X|_U)$  is called an *open subscheme* of  $X$ . If moreover  $U$  is an affine scheme, then we say that  $U$  is an *affine open subset/subscheme* of  $X$ . Observe that by lemma 2.22, affine open subsets of  $X$  form a basis of the topology of  $X$ .

(ii) Consider the natural morphism  $i_U : U \rightarrow X$  of schemes, where  $(i_U^\#)_V : \mathcal{O}_X(V) \rightarrow i_{U*}\mathcal{O}_X|_U(V) = \mathcal{O}_X(U \cap V)$  is the restriction. An *open immersion* is a morphism of schemes  $Y \rightarrow X$  which induces an isomorphism of  $Y$  with some open subscheme  $U \subseteq X$ , that is, it factors through  $i_U$ .

(iii) Let  $f : X \rightarrow Y$  be a morphism of schemes and  $U \subseteq X$  be open. Then we consider the restriction of  $f$  on  $U$ ,  $f|_U := f \circ i_U : U \rightarrow Y$ .

**Proposition 2.25** (Glueing of morphisms). Let  $X, Y$  be schemes and  $X = \bigcup_{i \in I} U_i$  be an open cover. Then the map

$$\text{Hom}_{\text{Sch}}(X, Y) \rightarrow \prod_{i \in I} \text{Hom}_{\text{Sch}}(U_i, Y), \quad f \mapsto (f|_{U_i})_i$$

is injective, and has image consisting of those  $(f_i)_i$  with  $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$  for all  $i, j \in I$ .

*Proof.* Let  $(f, f^\#) : X \rightarrow Y$  be given. It is clear that  $f$  can be recovered uniquely from the  $(f|_{U_i})_i$ . Let  $V \subseteq Y$  be open. Since  $f_V^\#(s) \in \mathcal{O}_X(f^{-1}(V))$  is uniquely determined by its restrictions to  $(f^{-1}(V) \cap U_i)_i$  by the sheaf condition, we can recover  $f^\#$  as well.

It is clear that any family  $(f_i)_i$  in the image satisfies the given condition. Conversely, suppose  $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$  holds for all  $i, j \in I$ . On the level of topological spaces the claim is clear: For  $x \in X$  set  $f(x) := f_i(x)$  for any  $i$  with  $x \in U_i$ . Now let  $U \subseteq Y$  be open,  $s \in \mathcal{O}_Y(U)$ . The family of sections  $t_i := (f_i)_U^\#(s) \in \mathcal{O}_{U_i}(U \cap U_i) = \mathcal{O}_X(U \cap U_i)$  glue to a unique section  $f_U^\#(s)$ , since  $t_i|_{U_i \cap U_j} = (f_i)_{U_i \cap U_j}^\#(s) = (f_j)_{U_i \cap U_j}^\#(s) = t_j|_{U_i \cap U_j}$  by assumption. By uniqueness, this construction respects addition and multiplication, so  $f_U^\#$  is a ring morphism. It is clear that  $f^\#$  is a morphism of sheaves, and that  $f|_{U_i} = f_i$ , hence  $f_x^\#$  is local for all  $x \in X$ , and  $(f, f^\#)$  is a morphism of schemes as required. □

### 3 Properties and some Constructions of Schemes

**Definition 3.1.** A scheme  $X$  is called *irreducible/connected* if its underlying topological space is.

**Definition 3.2.** A scheme  $X$  is called *reduced/integral* if for every open  $U \subseteq X$ , the ring  $\mathcal{O}_X(U)$  is reduced/an integral domain.

**Remark 3.3.** An affine scheme  $X = \operatorname{Spec} A$  is irreducible if and only if the nilradical of  $A$  is prime. In particular, if  $A$  is an integral domain then  $X$  is irreducible. Furthermore,  $X$  is reduced/integral if and only if  $A$  is.

**Remark 3.4.** Let  $X$  be a scheme. Then  $X$  is reduced if and only if  $\mathcal{O}_{X,x}$  is reduced for every  $x \in X$ . If  $X$  is integral, then so are the stalks, but the converse fails in general.

Lecture 10  
May 18, 2026

**Proposition 3.5.** *Let  $X$  be a scheme. Then  $X$  is integral if and only if  $X$  is reduced and irreducible.*

**Remark 3.6.** In the affine case  $X = \operatorname{Spec} A$ , this follows from remark 3.3 and lemma 2.7.

**Lemma 3.7.** *Let  $X$  be a scheme and  $f \in \mathcal{O}_X(X)$ . Define  $X_f := \{x \in X \mid f_x \notin \mathfrak{m}_x\}$ , where  $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$  is the maximal ideal of the local ring  $\mathcal{O}_{X,x}$ .*

- (i) *Let  $U = \operatorname{Spec} B \subseteq X$  be open and  $\tilde{f} := f|_U \in \mathcal{O}_X(U) \cong B$ . Then  $X_f \cap U = D(\tilde{f})$ .*
- (ii)  *$X_f$  is open in  $X$ .*

*Proof.* (ii) follows from (i), since  $X$  can be covered by affine opens. Let  $x \in U = \operatorname{Spec} B$ . Then  $f_x = \tilde{f}_x$ , so  $x \in X_f \cap \operatorname{Spec} B$  if and only if  $\tilde{f}_x = \frac{\tilde{f}}{1} \notin \mathfrak{m}_x = xB_x$ , if and only if  $\tilde{f} \notin x$ , i.e.  $x \in D(\tilde{f})$ .  $\square$

*Proof of Proposition 3.5.* First let  $X$  be integral. Then it is clear that  $X$  is reduced. Let  $\emptyset = U_1 \cap U_2$  with  $U_1, U_2$  open. By the sheaf property,  $\mathcal{O}_X(U_1 \cup U_2) \cong \mathcal{O}_X(U_1) \times \mathcal{O}_X(U_2)$  is an isomorphism. Since  $\mathcal{O}_X(U_1 \cup U_2)$  is integral by assumption, we must have  $\mathcal{O}_X(U_1) = 0$ , say, and hence  $U_1 = \emptyset$ .

Conversely, assume that  $X$  is reduced and irreducible. Let  $U \subseteq X$  be open and  $f, g \in \mathcal{O}_X(U)$  with  $fg = 0$ . Consider the open subsets  $U_f, U_g \subseteq U$  from lemma 3.7. Then for  $x \in U$ , we have  $f_x g_x = 0 \in \mathfrak{m}_x$ , so  $f_x \in \mathfrak{m}_x$  or  $g_x \in \mathfrak{m}_x$ . Hence  $U_f \cap U_g = \emptyset$ . Since  $X$  is irreducible, so is  $U$ , and wlog it follows that  $U_f = \emptyset$ . Showing that  $f = 0$  can be done on an affine cover, so wlog  $U = \operatorname{Spec} B$  is affine. Then lemma 3.7 says  $D(f) = \emptyset$ , and lemma 2.2 yields  $f \in \operatorname{Nil} B = 0$ .  $\square$

**Proposition 3.8.** *Let  $X$  be an irreducible scheme. Then there exists a unique so-called generic point  $\xi \in X$  such that  $\overline{\{\xi\}} = X$ . If  $X$  is integral, then  $\mathcal{O}_{X,\xi}$  is a field, called its function field.*

*Proof.* If  $X = \operatorname{Spec} A$  is affine, then  $\overline{\{x\}} = V(x)$  by remark 2.5, and  $\operatorname{Spec} A = V(0)$  is equivalent to  $x = \operatorname{Nil} A$  by lemma 2.2. If  $A$  is integral, then  $\xi = \operatorname{Nil} A = 0$ , so  $\mathcal{O}_{X,\xi} \cong A_{(0)} = \operatorname{Frac} A$  is a field.

For the general case, assume that  $\xi$  is a generic point. Then  $\xi|_{\operatorname{Spec} A} = \operatorname{Nil} A$  for any affine open  $\operatorname{Spec} A \subseteq X$ , so  $\xi$  is uniquely determined by the sheaf condition. For existence, let  $\xi$  be the generic point of any affine open  $U = \operatorname{Spec} A \subseteq X$ , and let  $V \subseteq X$  be open. Since  $X$  is irreducible,  $U \cap V$  is a non-empty open in  $U$ , so contains  $\xi$ . Hence  $\xi \in V$  for any  $V \subseteq X$  open, i.e.  $\overline{\{\xi\}} = X$ .

Now let  $X$  be integral, and let  $U = \operatorname{Spec} A \subseteq X$ . Then by the previous paragraph,  $0 \subseteq A$  is the generic point  $\xi$  of  $X$ , and  $\mathcal{O}_{X,\xi} = \mathcal{O}_{U,0} = \operatorname{Frac} A$  is a field.  $\square$

**Remark 3.9.** The proof shows: If  $X$  is integral with function field  $K$ , then  $\text{Frac } A \cong K$  for every affine open  $\text{Spec } A \subseteq X$ . In particular, all affine opens in  $X$  have the same field of fractions.

**Definition 3.10.** A topological space  $X$  is *Noetherian* if  $X$  satisfies the descending chain condition for closed subsets, that is: Each chain  $X \supseteq X_1 \supseteq X_2 \supseteq \dots$  with  $X_i$  closed stabilizes, i.e. there is  $m \geq 1$  with  $X_m = X_n$  for all  $n \geq m$ .

**Proposition 3.11.** *If  $A$  is a Noetherian ring, then  $\text{Spec } A$  is Noetherian.*

**Remark 3.12.** The converse is not true in general. For example, let  $k$  be an algebraically closed field and take  $A = k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$ .

*Proof.* Let  $V(I_1) \supseteq V(I_2) \supseteq \dots$  be a descending chain of closed subsets in  $\text{Spec } A$ . Since  $V(J) = V(\sqrt{J})$  for any ideal  $J$ , wlog all the  $I_k$  are radical. Then by lemma 2.2, we get reverse inclusions  $I_1 \subseteq I_2 \subseteq \dots$ , which stabilize by assumption.  $\square$

Lecture 11  
May 21, 2026

**Definition 3.13.** A topological space  $X$  is called *quasi-compact* if every open cover of  $X$  has a finite subcover. A scheme is quasi-compact if its underlying topological space is quasi-compact.

In some domains, quasi-compactness is usually just called compactness. In a more general setting, one sets compactness to mean quasi-compact and Hausdorff, which is why the difference rarely appears in fields related to Euclidean topologies. But the Zariski topology is almost never Hausdorff, so we use the term quasi-compactness to highlight this difference.

**Lemma 3.14.** *Let  $X = \text{Spec } A$  be an affine scheme. Then  $X$  is quasi-compact. Therefore, a scheme  $X$  is quasi-compact if and only if  $X$  can be covered by finitely many affine open subschemes.*

*Proof.* Let  $X = \bigcup_{i \in I} U_i$  with  $U_i$  open. Let  $X \setminus U_i = V(I_i)$ , then  $\emptyset = \bigcap_i V(I_i) = V(\sum_i I_i)$ , which is equivalent to  $1 \in \sum_i I_i$ . But this means that 1 can be written as a *finite* linear combination of elements in the  $I_i$ , so those finitely many ideals suffice.  $\square$

**Definition 3.15.** (i) A scheme  $X$  is *locally Noetherian* if there is an affine open cover  $X = \bigcup_{i \in I} \text{Spec } A_i$  such that all  $A_i$  are Noetherian rings.  
(ii)  $X$  is *Noetherian* if there is a finite open cover  $X = \bigcup_{i=1}^n \text{Spec } A_i$  such that all  $A_i$  are Noetherian rings. Equivalently,  $X$  is Noetherian if it is locally Noetherian and quasi-compact.

**Proposition 3.16.** *If  $X$  is a Noetherian scheme, then its underlying space is Noetherian.*

*Proof.*  $X = \bigcup_{i=1}^n \text{Spec } A_i$  is a finite union of Noetherian topological spaces by proposition 3.11. Finite unions of Noetherian spaces are again Noetherian.  $\square$

**Proposition 3.17.** (i) *A topological space  $X$  is Noetherian if and only if every open  $U \subseteq X$  is quasi-compact.*

(ii) *Let  $X$  be a Noetherian scheme and  $U \subseteq X$  be open. Then  $U$  is also a Noetherian scheme.*

*Proof.* Exercise.  $\square$

**Proposition 3.18 (Nike's Lemma).** *Let  $\text{Spec } A, \text{Spec } B$  be two affine open subschemes of a scheme  $X$ . Then  $\text{Spec } A \cap \text{Spec } B$  can be covered by open subsets which are simultaneously distinguished opens in  $\text{Spec } A$  and  $\text{Spec } B$ .*

*Proof.* Let  $x \in \operatorname{Spec} A \cap \operatorname{Spec} B$  and let  $x \in D(h) \subseteq \operatorname{Spec} A \cap \operatorname{Spec} B$  be a principal neighbourhood in  $\operatorname{Spec} A$ . By the exercises,  $D(h) \cong \operatorname{Spec} A_h \subseteq \operatorname{Spec} B$ . Then find a distinguished open  $x \in \operatorname{Spec} B_g \subseteq \operatorname{Spec} A_h$ . These inclusions of affine schemes are induced by ring morphisms  $\varphi : B \rightarrow A_h$  and  $\psi : A_h \rightarrow B_g$  with  $\psi \circ \varphi : B \rightarrow B_h$  being the canonical localization map. By sheet 4, exercise 1,  $D(g) = D(\varphi(g))$  is distinguished in  $\operatorname{Spec} A_h$ , hence also in  $\operatorname{Spec} A$ , since if  $\varphi(g) = \frac{a}{h^m}$ , then  $D(\varphi(g)) = D(h) \cap D(a) = D(ha)$   $\square$

**Lemma 3.19** (Affine Communication Lemma). *Let  $X$  be a scheme and  $\mathcal{P}$  be a property of open affine subschemes of  $X$ , such that*

- (i) *if  $\operatorname{Spec} A \hookrightarrow X$  has  $\mathcal{P}$ , then so does every  $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A \hookrightarrow X$ , and*
- (ii) *if  $(f_1, \dots, f_n) = A$  and  $\operatorname{Spec} A_{f_i} \hookrightarrow \operatorname{Spec} A \hookrightarrow X$  have  $\mathcal{P}$ , then so does  $\operatorname{Spec} A$ .*

*Assume  $X = \bigcup_{i \in I} \operatorname{Spec} A_i$  with all  $\operatorname{Spec} A_i$  satisfying  $\mathcal{P}$ . Then every affine open subset of  $X$  satisfies  $\mathcal{P}$ .*

*Proof.* Let  $\operatorname{Spec} A \subseteq X$  be an affine open. By Proposition 3.18 we can write  $\operatorname{Spec} A = \bigcup_{j \in J} U_j$  such that  $U_j$  is distinguished both in  $\operatorname{Spec} A$  and in some  $\operatorname{Spec} A_i$ . By (i) all  $U_j$  satisfy  $\mathcal{P}$ . Since  $\operatorname{Spec} A$  is quasi-compact by lemma 3.14, we can find a finite subcover, and the result follows from (ii).  $\square$

**Proposition 3.20.** *Let  $X$  be a scheme. Then  $X$  is locally Noetherian if and only if for every affine open  $\operatorname{Spec} A \subseteq X$ ,  $A$  is a Noetherian ring.*

**Corollary 3.21.**  *$\operatorname{Spec} A$  is a Noetherian scheme if and only if  $A$  is Noetherian.*

*Proof (of Proposition 3.20).* We have to check conditions (i),(ii) of Lemma 3.19 for the property  $\mathcal{P}(\operatorname{Spec} A) = A$  is Noetherian. Since every localization of a Noetherian ring is again Noetherian, (i) is clear. So let  $(f_1, \dots, f_n) = A$  and assume that all  $A_{f_i}$  are Noetherian. Write  $\varphi_i : A \rightarrow A_{f_i} =: S_i^{-1}A$  for the canonical localization maps. Let  $I \subseteq A$  be an ideal, then  $I = \bigcap_{i=1}^n \varphi_i^{-1}(S_i^{-1}I)$ : Indeed, " $\subseteq$ " is clear, so assume  $a \in \bigcap_{i=1}^n \varphi_i^{-1}(S_i^{-1}I) = \bigcap_{i=1}^n I^{S_i}$ . It follows that  $f_i^{k_i}a \in I$  for suitable  $k_1, \dots, k_n \in \mathbb{N}$ . Since  $1 = (f_1, \dots, f_n) = (f_1^{k_1}, \dots, f_n^{k_n})$ , we get  $a \in (f_1^{k_1}a, \dots, f_n^{k_n}a) \subseteq I$ .

Let  $I_1 \subseteq I_2 \subseteq \dots$  be an ascending chain of ideals in  $A$ . For every  $i = 1, \dots, n$ , the localized chain  $S_i^{-1}I_1 \subseteq S_i^{-1}I_2 \subseteq \dots$  stabilizes by assumption. Once these finally many chains have all stabilized, by the above formula so does the original chain.  $\square$

Let  $A$  be a ring. A scheme over  $A$  (or  $A$ -scheme) is a scheme  $X$  with a morphism  $X \rightarrow \operatorname{Spec} A$ . An affine open subset  $\operatorname{Spec} B \subseteq X$  thus induces a ring map  $A \rightarrow B$ , so  $B$  can be considered an  $A$ -algebra.

**Definition 3.22.** Let  $X$  be a scheme over  $A$ .

- (i)  $X$  is *locally of finite type* over  $A$  if  $X = \bigcup_i \operatorname{Spec} B_i$ , such that  $B_i$  is a finitely generated  $A$ -algebra.
- (ii)  $X$  is of *finite type* over  $A$  if  $X$  is locally of finite type and quasi-compact, equivalently if  $X = \bigcup_{i=1}^n \operatorname{Spec} B_i$  with all  $B_i$  finitely generated  $A$ -algebras.

**Proposition 3.23.** *A scheme  $X$  is locally of finite type over  $A$  if and only if for every affine open  $\operatorname{Spec} B \subseteq X$ ,  $B$  is a finitely generated  $A$ -algebra.*

*Proof.* Exercise (use lemma 3.19 again).  $\square$

**Corollary 3.24.**  *$\operatorname{Spec} B$  is of finite type over  $A$  if and only if  $B$  is a finitely generated  $A$ -algebra.*

Lecture 12  
May 28, 2026

**Definition 3.25.** Let  $K$  be a field. An affine scheme  $X$  which is reduced and of finite type over  $K$  is called an *affine variety* over  $K$ .

In the literature, for an affine variety over  $K$  sometimes it is required that  $K$  is algebraically closed. A general variety over  $K$  can be defined as a reduced scheme of finite type over  $K$ . A more common definition also requires the scheme to be separated, more on this later.

First we want to have a notion of projective spaces in our language of schemes. Let  $B = \bigoplus_{i \geq 0} B_i$  be a graded ring. Elements of some  $B_i$  are called homogenous elements of degree  $i$ . Set  $B_+ := \bigoplus_{i > 0} B_i$ . An ideal  $I \subseteq B$  is called homogeneous if it is generated by homogeneous elements, equivalently if  $b = b_0 + b_1 + \dots + b_n$  is the decomposition into a sum of graded components, then  $b \in I$  if and only if  $b_i \in I$  for all  $i$ . A homogeneous ideal  $\mathfrak{p} \subseteq B$  is prime if and only if  $bc \in \mathfrak{p} \Rightarrow b \in \mathfrak{p}$  or  $c \in \mathfrak{p}$  for all homogeneous  $b, c \in B$ .

Let  $\text{Proj } B$  be the set of all homogeneous prime ideals  $\mathfrak{p} \subseteq B$  with  $B_+ \not\subseteq \mathfrak{p}$ . For a homogeneous ideal  $I \subseteq B$ , set  $V_+(I) := \{\mathfrak{p} \in \text{Proj } B \mid I \subseteq \mathfrak{p}\}$ . As in the affine case, we have

**Lemma 3.26.** If  $I, J, (I_\alpha)_{\alpha \in A}$  are homogeneous ideals, then  $V_+(IJ) = V_+(I) \cup V_+(J)$  and  $V_+(\sum_\alpha I_\alpha) = \bigcap_\alpha V_+(I_\alpha)$ .  $\square$

By lemma 3.26 we can define the *Zariski topology* on  $\text{Proj } B$  by declaring closed subsets to be the subsets of the form  $V_+(I)$  for homogeneous ideals  $I \subseteq B$ .

**Remark 3.27.** Let  $I = (\{f_j\}_j)$  be a homogeneous ideal, with a homogeneous generating set  $\{f_j\}_j$ . Then  $V_+(I) = \emptyset$  if and only if  $\text{Proj } B = \bigcup_j D_+(f_j)$  if and only if  $B_+ \subseteq \sqrt{I}$  (Exercise).

**Definition 3.28.** For any homogeneous element  $f \in B_+$  we define  $D_+(f) = \{\mathfrak{p} \in \text{Proj } B \mid f \notin \mathfrak{p}\} = \text{Proj } B \setminus V_+(\{f\})$ . As in the affine case, we have  $D_+(f) \cap D_+(g) = D_+(fg)$  and the  $\{D_+(f)\}_f$  form a basis of the topology on  $\text{Proj } B$ .

**Remark 3.29.** Let  $\mathfrak{p} \in \text{Proj } B$ . Let  $T_{\mathfrak{p}} \subseteq B$  be the set of homogeneous elements not in  $\mathfrak{p}$ . Then  $T_{\mathfrak{p}}^{-1}$  is a  $(\mathbb{Z})$ -graded ring with  $\deg(\frac{a}{s}) := \deg a - \deg s$ . Set  $B_{(\mathfrak{p})} \subseteq T_{\mathfrak{p}}^{-1}B$  to be the subring of elements of degree 0, i.e.  $B_{(\mathfrak{p})} = \{\frac{a}{s} \mid a, s \text{ homogeneous, } s \notin \mathfrak{p} \text{ and } \deg(a) = \deg(s)\}$ .  $B_{(\mathfrak{p})}$  is a local ring with maximal ideal  $\mathfrak{n}_{(\mathfrak{p})} := B_{(\mathfrak{p})} \cap \mathfrak{p}T_{\mathfrak{p}}^{-1}B$ , for if  $\frac{a}{b} \in B_{(\mathfrak{p})} \setminus \mathfrak{n}_{(\mathfrak{p})}$ , then  $a \notin \mathfrak{p}$  and  $\frac{b}{a}$  still has degree 0, so  $\frac{b}{a} \in B_{(\mathfrak{p})}$  and  $\frac{a}{b}$  is a unit.

**Definition 3.30.** Let  $X = \text{Proj } B$ . We define the *structure sheaf* of  $X$  as

$$\mathcal{O}_X(U) := \left\{ s : U \rightarrow \prod_{\mathfrak{p} \in U} B_{(\mathfrak{p})} \mid s(\mathfrak{p}) \in B_{(\mathfrak{p})} \text{ for all } \mathfrak{p} \in U, \text{ and } (*) \right\},$$

where  $(*)$  is the condition that for any  $\mathfrak{q} \in U$  there exists an open neighbourhood  $V \subseteq U$  and homogeneous elements  $f, g \in B$  of the same degree such that  $g \notin \mathfrak{p}$  and  $s(\mathfrak{p}) = \frac{f}{g} \in B_{(\mathfrak{p})}$  for all  $\mathfrak{p} \in V$ .

Clearly  $(X, \mathcal{O}_X)$  is a ringed space. By the same proof as in proposition 2.11,  $\mathcal{O}_{X, \mathfrak{p}} \cong B_{(\mathfrak{p})}$ , so by remark 3.29  $(X, \mathcal{O}_X)$  is a locally ringed space.

**Theorem 3.31.**  $(X, \mathcal{O}_X)$  is a scheme. More precisely, for every  $f \in B_+$  we have an isomorphism  $(D_+(f), \mathcal{O}_{D_+(f)}) \cong (\text{Spec } B_{(f)}, \mathcal{O}_{\text{Spec } B_{(f)}})$ , where  $B_{(f)} \subseteq B_f$  is the subring of elements of degree 0.

**Example 3.32.** Let  $B = K[x_0, \dots, x_n]$  with the usual grading,  $X = \mathbb{P}_K^n := \text{Proj } B$  and  $f = x_i$ . Then  $B_{(x_i)} \cong k[x_0, \dots, \hat{x}_i, \dots, x_n]$  via inhomogenization, so theorem 3.31 says that  $\mathbb{P}_K^n$  is covered by copies of  $\mathbb{A}_K^n \cong D_+(x_i)$ , in accordance with the classical situation.

*Proof of Theorem 3.31.* Note that  $\text{Proj } B$  is a subspace of  $\text{Spec } B$ , since if  $f = f_0 + \dots + f_r \in B$ , then  $D(f) \cap \text{Proj } B = \bigcup_i D_+(f_i)$ . In particular, if  $f$  is homogeneous, then  $D(f) \cap \text{Proj } B = D_+(f)$ , so we get a continuous map  $\theta : D_+(f) \hookrightarrow D(f) \cong \text{Spec } B_f \rightarrow \text{Spec } B_{(f)}$ ,  $\mathfrak{p} \mapsto \mathfrak{p}B_f \cap B_{(f)}$ . We claim that  $\theta$  is a homeomorphism.  $\theta$  is open, since if  $D_+(g) \subseteq D_+(f)$ , then  $\theta(D_+(g)) = D(h)$  with  $h = \frac{g^{\deg f}}{f^{\deg g}}$ . (Exercise.) So it remains to show that  $\theta$  is bijective.

Let  $d := \deg f$ . For  $\mathfrak{q} \in \text{Spec } B_{(f)}$  set  $\psi(\mathfrak{q}) := \bigoplus_{n \geq 0} \psi(\mathfrak{q})_n$ , where  $\psi(\mathfrak{q})_n := \{b \in B_n \mid \frac{b^d}{f^n} \in \mathfrak{q}\}$ .  $\psi(\mathfrak{q})$  is a graded ideal in  $B$ : Let  $a \in B_m$  and  $b \in \psi(\mathfrak{q})_n$ , then  $\frac{(ab)^d}{f^{m+n}} = \frac{a^d}{f^m} \frac{b^d}{f^n} \in \mathfrak{q}$ , so  $ab \in \psi(\mathfrak{q})_{m+n}$ . Also if  $a, b \in \psi(\mathfrak{q})_n$ , then  $(\frac{(a+b)^d}{f^n})^2 = \frac{1}{f^{2d}} \sum_i \binom{2d}{i} a^i b^{2d-i} \in \mathfrak{q}$ , and since  $\mathfrak{q}$  is prime,  $\frac{(a+b)^d}{f^n} \in \mathfrak{q}$ , i.e.  $a + b \in \psi(\mathfrak{q})_n$ . Furthermore,  $f \notin \psi(\mathfrak{q})$ , since otherwise by construction  $\frac{f^d}{f^d} = 1 \in \mathfrak{q}$ . To see that  $\psi(\mathfrak{q})$  is prime, let  $a \in B_m$ ,  $b \in B_n$  with  $ab \in \psi(\mathfrak{q})_{n+m}$ . That is,  $\frac{a^d b^d}{f^{m+n}} \in \mathfrak{q}$ , so  $\frac{a^d}{f^m} \in \mathfrak{q}$  or  $\frac{b^d}{f^n} \in \mathfrak{q}$ , which exactly means  $a \in \psi(\mathfrak{q})_m$  or  $b \in \psi(\mathfrak{q})_n$ . Thus we have successfully constructed a map  $\psi : \text{Spec } B_{(f)} \rightarrow D_+(f)$  in the inverse direction, and one checks that it is indeed inverse to  $\theta$  (Exercise).

It remains to construct an isomorphism  $\theta^\sharp : \mathcal{O}_{\text{Spec } B_{(f)}} \rightarrow \theta_* \mathcal{O}|_{D_+(f)}$ . For this, note that for  $\mathfrak{p} \in D_+(f)$ , we have natural isomorphisms  $\varphi_{\mathfrak{p}} : (B_{(f)})_{\theta(\mathfrak{p})} \rightarrow B_{(\mathfrak{p})}$ ,  $\frac{a/f^n}{b/f^m} \mapsto \frac{af^m}{bf^n}$  from universal properties of localization. Indeed,  $\frac{b}{f^n} \notin \theta(\mathfrak{p})$  if and only if  $b \notin \mathfrak{p}$  for  $b \in B_{dn}$ , injectivity of  $\varphi_{\mathfrak{p}}$  is clear, and for  $\frac{a}{b} \in B_{(f)}$  one has  $\varphi_{\mathfrak{p}}(\frac{ab^{d-1}/f^{\deg a}}{b^d/f^{\deg b}}) = \frac{a}{b}$ . Hence we can identify the sections of the two sheaves, i.e. define  $\theta^\sharp_{\theta(U)} : \mathcal{O}_{\text{Spec } B_{(f)}}(\theta(U)) \rightarrow \mathcal{O}_{\text{Proj } B}(U)$ ,  $s \mapsto (\mathfrak{p} \mapsto \varphi_{\mathfrak{p}}(s(\theta(\mathfrak{p}))))$ , which is clearly an isomorphism for any  $U \subseteq D_+(f)$  open.  $\square$

Lecture 14  
Jun 8, 2026

**Example 3.33.** (i) Let  $B$  be a graded  $R$ -algebra, i.e.  $\varphi : R \rightarrow B$  with  $\varphi(R) \subseteq B_0$ . Then  $\text{Proj } B$  is a scheme over  $R$ , since all localizations  $B_{(\mathfrak{p})}$  are  $R$ -algebras.  
(ii) For any ring  $R$  set  $\mathbb{P}_R^n := \text{Proj } R[x_0, \dots, x_n]$  with the usual grading (cf. example 3.32), which is a scheme over  $R$  by (i). Clearly,  $\mathbb{P}_R^n$  is a scheme of finite type over  $R$ .

**Definition 3.34.** (i) Let  $s_X : X \rightarrow \text{Spec } R$ ,  $s_Y : Y \rightarrow \text{Spec } R$  be schemes over a ring  $R$ . A *morphism of  $R$ -schemes* is a morphism of schemes  $f : X \rightarrow Y$  compatible with the structure maps, i.e.  $s_X = s_Y \circ f$ . In this way,  $R$ -schemes form a category, with terminal object  $\text{id} : \text{Spec } R \rightarrow \text{Spec } R$ .  
(ii) Let  $X$  be a scheme over  $R$ . The  $R$ -points of  $X$  are  $X(R) := \text{Hom}_{R\text{-Sch}}(\text{Spec } R, X)$ . A morphism  $f : X \rightarrow Y$  of  $R$ -schemes induces a map  $X(R) \rightarrow Y(R)$  of sets.  
(iii) Let  $X$  be a scheme and  $x \in X$ . The *residue field* of  $x$  is  $k(x) := \mathcal{O}_{X,x}/\mathfrak{m}_x$ .

If  $k$  is a field, and  $X$  is a scheme over  $k$ , then  $X(k) = \{x \in X \mid k(x) = k\}$  (Exercise). If  $X = \text{Spec } k[X_1, \dots, X_n]/(f_1, \dots, f_m)$ , there is a bijection  $X(k) \cong \{\alpha \in k^n \mid f_i(\alpha) = 0 \text{ for all } i = 1, \dots, m\}$  (Exercise), i.e. we can “recover” the classical affine variety from  $X$ . The same holds in the projective case:

**Remark 3.35.** (i) Let  $k$  be a field.  $\mathbb{P}_k^n(k) \cong (k^{n+1} \setminus \{0\})/(x \sim \lambda x)$  can be identified with the underlying set of classical projective space (Exercise). As above, one can consider more generally  $X = \text{Proj } k[X_0, \dots, X_n]/(f_1, \dots, f_m)$  (with  $f_i$  graded), then  $X(k) = \{x = [x_0 : \dots : x_n] \mid f_1(x) = \dots = f_m(x) = 0\}$  under the same identification.  
(ii) If  $B$  is integral, then  $\text{Proj } B$  is an integral scheme. In particular, if  $I \subseteq k[X_0, \dots, X_n]$  is prime, then  $\text{Proj } k[X_0, \dots, X_n]/I$  is an integral noetherian scheme of finite type over  $k$ .

Lecture 15  
Jun 11, 2026

**Proposition 3.36.** Let  $\varphi : C \rightarrow B$  be a homomorphism of graded rings (i.e. there is  $r \geq 1$  such that  $\varphi(C_d) \subseteq B_{rd}$  for all  $d$ ). Consider the ideal  $J = \langle \varphi(C_+) \rangle$ . Then  $\varphi$  induces a morphism



of schemes  $f : \text{Proj } B \setminus V_+(J) \rightarrow \text{Proj } C$ ,  $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ . Moreover, for  $h \in C_+$  we have  $f^{-1}(D_+(h)) = D_+(\varphi(h))$  and

$$\begin{array}{ccc} D_+(\varphi(h)) & \xrightarrow{f} & D_+(h) \\ \downarrow \cong & & \downarrow \cong \\ \text{Spec } B_{(\varphi(h))} & \xrightarrow{\text{Spec } \bar{\varphi}} & \text{Spec } C_{(h)} \end{array}$$

commutes, where the vertical arrows are the isomorphisms from proposition 3.31 and  $\bar{\varphi} : C_{(h)} \rightarrow B_{(\varphi(h))}$  is the map induced from  $\varphi$  by localization.

*Proof.* Exercise (similar to the affine case). □

**Example 3.37.** (i) Let  $\varphi : K[X_0, X_1] \rightarrow K[X_0, X_1, X_2]$  be the inclusion. In the notation of proposition 3.35,  $J = (X_0, X_1) \subseteq K[X_0, X_1, X_2]$  (corresponding to the  $K$ -rational point  $(0 : 0 : 1)$ ), so we get a morphism  $\mathbb{P}_K^3 \setminus \{J\} \rightarrow \mathbb{P}_K^2$ . On  $K$ -rational points it is given as  $[a : b : c] \mapsto [a : b]$ .

(ii) Let  $\varphi : K[X_0, X_1, X_2] \rightarrow K[X_0, X_1]$  be given by  $X_2 \mapsto 0$ . Then  $J = B_+$ , so we obtain a map  $\mathbb{P}_K^2 \rightarrow \mathbb{P}_K^3$ , which on  $K$ -points is  $[a : b] \mapsto [a : b : 0]$ .

## 4 Closed Subschemes and Fibred Products of Schemes

**Example 4.1.** Let  $A$  be a ring,  $I \subseteq A$  be an ideal. Then  $A \rightarrow A/I$  induces a morphism of schemes  $f : \operatorname{Spec} A/I \rightarrow \operatorname{Spec} A$ , with image  $V(I)$ , which is closed.

**Remark 4.2.** Closed subsets of  $\operatorname{Spec} A$  are of the form  $V(I)$ . By example 4.1, we can define a scheme structure on  $V(I)$ , namely  $(V(I), f_* \mathcal{O}_{\operatorname{Spec} A/I}) \cong \operatorname{Spec} A/I$ . However, for another ideal  $J \neq I$  we may have  $V(I) = V(J)$  and we get another scheme structure on  $V(I)$ .

**Remark 4.3.** A closed immersion (or closed embedding) should intuitively be a morphism  $f : Z \rightarrow X$  of schemes such that  $f$  induces a homeomorphism  $f : Z \rightarrow f(Z) \subseteq X$  with  $f(Z)$  closed in  $X$ , and if  $X = \bigcup_i U_i$  is an open affine cover,  $U_i = \operatorname{Spec} A_i$ , then  $f^{-1}(U_i) \cong \operatorname{Spec} A_i/I$  for some ideal  $I \subseteq A_i$ , and  $f|_{f^{-1}(U_i)}$  looks like  $\operatorname{Spec} A_i/I \rightarrow \operatorname{Spec} A$ . For practical purposes, we will give an a priori different definition, which will turn out to be equivalent:

**Definition 4.4.** A *closed immersion* is a morphism  $f : Z \rightarrow X$  of schemes such that  $f$  induces a homeomorphism  $f : Z \rightarrow f(Z)$  with  $f(Z) \subseteq X$  closed, and  $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Z$  is surjective.

**Example 4.5.** Let  $I \subseteq A$  be an ideal. The morphism  $f : \operatorname{Spec} A/I \rightarrow \operatorname{Spec} A$  from example 4.1 is a closed immersion.

**Definition 4.6.** A *closed subscheme* of a scheme  $X$  is an equivalence class of closed immersions  $f : Z \rightarrow X$ , where  $f$  is equivalent to  $f' : Z' \rightarrow X$  if and only if there is an isomorphism  $g : Z \rightarrow Z'$  with  $f = f' \circ g$ .

**Remark 4.7.** Equivalently, a closed subscheme of  $X$  is given by a closed subset  $Z \subseteq X$  with a scheme structure  $(Z, \mathcal{O}_Z)$  such that there is a closed immersion  $Z \rightarrow X$  which is the inclusion on the level of topological spaces.

**Remark 4.8.** In the affine case,  $\operatorname{Spec} A/I \rightarrow \operatorname{Spec} A$  is a closed subscheme.

**Proposition 4.9.** Let  $X = \operatorname{Spec} A$  be an affine scheme and  $f : Z \rightarrow X$  be a closed immersion. Then there exists a unique ideal  $I \subseteq A$  such that  $f : Z \rightarrow X$  is equivalent to  $\operatorname{Spec} A/I \rightarrow \operatorname{Spec} A$  in the sense of definition 4.6.

**Corollary 4.10.**  $f : Z \rightarrow X$  is a closed immersion if and only if for every affine open  $\operatorname{Spec} A \cong U \subseteq X$ , the restriction  $f|_{f^{-1}(U)} : f^{-1}(U) \rightarrow \operatorname{Spec} A$  is equivalent to  $\operatorname{Spec} A/I \rightarrow \operatorname{Spec} A$  for some ideal  $I \subseteq A$ , as closed immersions to  $\operatorname{Spec} A$ .

**Remark 4.11.** Let  $X$  be a scheme and  $f \in \mathcal{O}_X(X)$ . Recall from lemma 3.7 the open set  $X_f = \{x \in X \mid f_x \notin \mathfrak{m}_x\} = \{x \in X \mid f_x \neq 0 \in k(x)\}$ . Let  $\varphi : X \rightarrow \operatorname{Spec} A$  be a morphism of schemes and  $g \in A$ . Then  $\varphi^{-1}(D(g)) = X_{\varphi^*_{\operatorname{Spec} A}(g)}$ . Indeed, this can be checked locally and is clear from the construction if  $X$  is affine (see sheet 4, exercise 1).

**Lemma 4.12.** Let  $X$  be a scheme and  $X = \bigcup_{i=1}^n U_i$  be a finite affine open cover such that  $U_i \cap U_j$  admits a finite affine open cover for every  $i, j$ . (For example if  $X$  is affine or Noetherian.) Then for  $f \in \mathcal{O}_X(X)$ , the restriction map  $\mathcal{O}_X(X) \rightarrow \mathcal{O}_X(X_f)$  induces an isomorphism  $\mathcal{O}_X(X)_f \rightarrow \mathcal{O}_X(X_f)$ .

*Proof.* Exercise. □

**Lemma 4.13** (Affineness criterion). Let  $X$  be a scheme. Assume that there exist  $f_1, \dots, f_r \in \mathcal{O}_X(X)$  such that  $(f_1, \dots, f_r) = \mathcal{O}_X(X)$  and  $X_{f_i}$  are affine for all  $i = 1, \dots, r$ . Then  $X$  is affine.

*Proof.* Note that  $X = \bigcup_{i=1}^r X_{f_i}$  since for every  $x \in X$ , the formula  $(f_1, \dots, f_r) = 1$  also holds in  $k(x)$ , so not all of them can vanish at  $x$ .

Since  $X_{f_i}$  is affine, so is  $X_{f_i} \cap X_{f_j}$  by lemma 3.7. Thus the requirements of lemma 4.12 are satisfied for this cover, and we get isomorphisms  $\mathcal{O}_X(X)_{f_i} \rightarrow \mathcal{O}_X(X_{f_i})$ . Write  $A := \mathcal{O}_X(X)$  and  $X_{f_i} = \text{Spec } A_{f_i}$ . Then the lemma identifies the restrictions  $A \rightarrow A_i$  with the localizations  $A \rightarrow A_{f_i}$ . By sheet 4, exercise 4 we get a morphism  $g : X \rightarrow \text{Spec } A$  corresponding to  $\text{id}_A$ . By the above, the compositions  $\text{Spec } A_{f_i} \cong X_{f_i} \rightarrow X \rightarrow \text{Spec } A$  are induced by the canonical localization maps  $A \rightarrow A_{f_i}$ .

By remark 4.11 we get  $g^{-1}(D(f_i)) = X_{f_i}$ , so  $g|_{X_{f_i}} : X_{f_i} \rightarrow D(f_i)$  is an isomorphism. It follows from  $\bigcup_{i=1}^r D(f_i) = \text{Spec } A$  and the lemma below that  $g$  is an isomorphism.  $\square$

**Lemma 4.14.** *Let  $f : X \rightarrow Y$  be a morphism of schemes and  $Y = \bigcup_i V_i$  an open cover. Then  $f$  is an isomorphism if and only if  $f|_{f^{-1}(V_i)} : f^{-1}(V_i) \rightarrow V_i$  are isomorphisms for all  $i$ .*

*Proof.* The claim is clear on the level of topological spaces. For the maps of sheaves, it follows from the fact that isomorphisms can be checked on stalks (by proposition 1.11).  $\square$

*Proof of Proposition 4.9.* Cover  $Z$  by affine opens  $\text{Spec } B_i \cong f^{-1}(U_i)$  for some  $U_i \subseteq \text{Spec } A$ . Let  $D(g_j) \subseteq U_i$ . Write  $\varphi = f_{\text{Spec } A}^\# : A \rightarrow \mathcal{O}_Z(Z)$ , then  $f^{-1}(D(g_j)) = X_{\varphi(g_j)} \subseteq \text{Spec } B_i$  is affine by remark 4.11 and lemma 3.7. Since  $Z \simeq f(Z)$  is quasi-compact (as closed subset of a quasi-compact space), finitely many such  $X_{\varphi(g_1)}, \dots, X_{\varphi(g_n)}$  cover  $Z$ . Now  $\text{Spec } A$  is covered by  $D(g_1), \dots, D(g_n)$  and some  $D(g'_k)$  that cover  $X \setminus f(Z)$ . Then  $(g_i, g'_k)_{i,k} = 1$  in  $A$  and hence  $(\varphi(g_i), \varphi(g'_k))_{i,k} = 1$  in  $\mathcal{O}_Z(Z)$ . Moreover,  $f^{-1}(D(g'_k)) = \emptyset$ , so  $\varphi(g'_k)$  is nilpotent (since the restriction to any affine open is). Therefore  $(\varphi(g_i))_i = 1$  and we can apply lemma 4.13 to deduce that  $Z$  is affine.

Write  $Z = \text{Spec } B$ , it remains to show that the corresponding map  $\psi : A \rightarrow B$  is surjective. Since  $\psi$  factors through  $A/\ker \psi$ ,  $f$  factors through  $\theta : \text{Spec } B \rightarrow \text{Spec } A/\ker \psi$ .

$$\begin{array}{ccc}
 B & \xleftarrow{\psi} & A \\
 & \searrow & \swarrow \\
 & A/J & 
 \end{array}
 \qquad
 \begin{array}{ccc}
 \text{Spec } B = Z & \xleftarrow{f} & \text{Spec } A \\
 & \searrow \theta & \swarrow \\
 & \text{Spec } A/\ker \psi & 
 \end{array}$$

It follows that  $\theta$  is injective (on topological spaces) and closed. By the following lemma,  $\text{im } \theta$  is dense, and hence  $\theta$  is a homeomorphism. The lemma also implies that  $\theta^\#$  is injective, and surjectivity follows from the diagram above and the definition of closed immersions.  $\square$

**Lemma 4.15.** *Let  $\varphi : A \rightarrow B$  be a ring homomorphism and  $(f, f^\#) : \text{Spec } B \rightarrow \text{Spec } A$  the corresponding morphism of schemes. Then  $\varphi$  is injective if and only if  $f^\#$  is injective. In this case,  $f$  is dominant, i.e.  $\text{im } f$  is dense in  $\text{Spec } A$ .*

*Proof.* Exercise.  $\square$

**Remark 4.16.** Let  $X$  be a scheme and  $Z \subseteq X$  be a closed subset. We want to introduce a scheme structure  $\mathcal{O}_Z$  on  $Z$  so that  $Z \rightarrow X$  becomes a closed immersion. In the affine case, let  $Z = V(I)$ . Then we may take  $\text{Spec } A/J$  for any  $J$  with  $Z = V(J)$ , so there may be different closed scheme structures. However, only  $J = \sqrt{I}$  yields a reduced subscheme.

The same is true in the general case:  $Z \subseteq X$  can be endowed with a unique reduced closed subscheme structure: Let  $X = \bigcup_i V_i$  with  $V_i \cong \text{Spec } A_i$ . Then  $Y_i := Z \cap V_i$  is closed in  $V_i$ , so  $Y_i = V(J_i)$  for a unique radical ideal  $J_i \subseteq A_i$ . Giving  $Y_i$  the reduced subscheme structure  $\text{Spec } A_i/J_i \rightarrow \text{Spec } A_i$  as above, one can check that these glue to a scheme structure on  $Z$  (cf. 4.21), which is still reduced by remark 3.4.

**Example 4.17.** Let  $K$  be a field,  $X = \mathbb{A}_K^2$ ,  $Z = \{(0,0)\} \subseteq X$ . Then  $J_1 = (X, Y)$  and  $J_2 = (X^2, Y)$  give two different closed subschemes  $Z_i := \text{Spec } K[X_1, X_2]/J_i \rightarrow X$ , with  $Z_1$  being the reduced one.

**Remark 4.18** (Intersection of closed subschemes). In the affine case  $X = \text{Spec } A$ , let  $Z_1 = \text{Spec } A/J_1$ ,  $Z_2 = \text{Spec } A/J_2$  be two closed subschemes. Since  $V(J_1) \cap V(J_2) = V(J_1 + J_2)$ , we define  $Z_1 \cap Z_2 = \text{Spec } A/(J_1 + J_2)$ . For example, the intersection of the parabola  $V(Y - X^2)$  and the  $X$ -axis  $V(Y) \subseteq \mathbb{A}_K^2$  is  $\text{Spec } K[X, Y]/(Y, Y - X^2) = \text{Spec } K[X, Y]/(Y, X^2) = \text{Spec } K[X]/(X^2)$ , so we can “see” that the intersection has multiplicity 2.

In general, if  $Z_1, Z_2 \subseteq X$  are two closed subschemes, then  $Z_1 \cap Z_2$  can be defined in a way that for every affine open  $\text{Spec } A \subseteq X$ , the scheme structure agrees with the construction above.

**Remark 4.19.** Let  $B = A[X_0, \dots, X_n]$  and  $X = \mathbb{P}_A^n = \text{Proj } B$ . Then every closed subscheme of  $X$  is of the form  $\text{Proj } A[X_0, \dots, X_n]/I$  for some homogeneous ideal  $B_+ \not\subseteq I \subseteq B$ .

**Proposition 4.20** (Glueing sheaves). *Let  $X$  be a topological space,  $X = \bigcup_i U_i$  an open cover. Assume the following glueing data is given:*

- (i) *A sheaf  $\mathcal{F}_i$  on every  $U_i$ ,*
- (ii) *For every  $i, j$  an isomorphism  $\varphi_{ij} : \mathcal{F}_i|_{U_i \cap U_j} \rightarrow \mathcal{F}_j|_{U_i \cap U_j}$  such that  $\varphi_{ii} = \text{id}_{\mathcal{F}_i}$  and  $\varphi_{ik}|_{U_i \cap U_k \cap U_j} = \varphi_{jk}|_{U_j \cap U_k \cap U_i} \circ \varphi_{ij}|_{U_i \cap U_j \cap U_k}$  for all  $i, j, k$ .*

*Then there exists a unique sheaf  $\mathcal{F}$  on  $X$  together with isomorphisms  $\psi_i : \mathcal{F}|_{U_i} \rightarrow \mathcal{F}_i$  such that  $\psi_j|_{U_i \cap U_j} = \varphi_{ij} \circ \psi_i|_{U_i \cap U_j}$ . We say that  $\mathcal{F}$  is obtained by glueing the sheaves  $\mathcal{F}_i$  along the  $\varphi_{ij}$ .*

*Proof.* Set  $\mathcal{F}(V) := \{(s_i)_i \in \mathcal{F}_i(U_i \cap V) \mid \varphi_{ij}(s_i|_{V \cap U_i \cap U_j}) = s_j|_{V \cap U_i \cap U_j}\}$ . One checks that this satisfies all requirements (Exercise).  $\square$

**Remark 4.21.** If  $(U_i, \mathcal{F}_i)$  are schemes and the  $\varphi_{ij}$  are morphisms of schemes, then  $(X, \mathcal{F})$  is a scheme as well.

**Proposition 4.22.** *Let  $(X_i)_{i \in I}$  be schemes. Given open subsets  $X_{ij} \subseteq X_i$  and isomorphisms  $\varphi_{ij} : X_{ij} \rightarrow X_{ji}$  of schemes for all  $i, j$  such that  $\varphi_{ij}(X_{ij} \cap X_{ik}) = X_{ji} \cap X_{jk}$  and  $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$  on  $X_{ij} \cap X_{ik}$  for all  $i, j, k$ .*

*Then there exists a unique scheme  $X$  with an open cover  $X = \bigcup U_i$  and isomorphisms  $\psi_i : U_i \rightarrow X_i$  of schemes such that  $\psi_j = \varphi_{ij} \circ \psi_i$  on  $U_i \cap U_j = \psi_i^{-1}(X_{ij})$ .*

*Proof.* Glue the topological spaces  $X_i$  along the  $\varphi_{ij}$  to a space  $X = \coprod X_i / \langle x_i \sim \varphi_{ij}(x_i) \rangle$ . Upon identifying the  $X_i$  with their homeomorphic image  $U_i$  in  $X$ , and transporting the sheaf structures, we are exactly in the situation of proposition 4.20.  $\square$

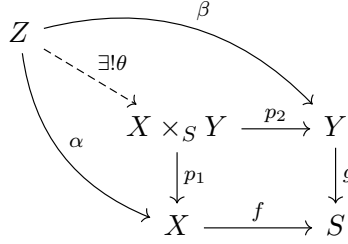
**Remark 4.23.** By proposition 2.25, a morphism  $f : X \rightarrow Y$  with  $X$  as above is uniquely determined by morphisms  $f_i : X_i \rightarrow Y$  such that  $f_i = f_j \circ \varphi_{ij}$  on  $X_{ij} \subseteq X_i$ .

**Example 4.24.** Let  $R$  be a ring. By example 3.32, we have  $\mathbb{P}_R^n = \bigcup D_+(x_i)$  with  $D_+(x_i) = \text{Spec } K[x_0, \dots, \widehat{x_i}, \dots, x_n] \cong \mathbb{A}_K^n$ . Hence  $\mathbb{P}_R^n$  could alternatively be obtained by glueing these  $\mathbb{A}_K^n$  along the morphisms induced by the ring maps

$$\varphi_{ij}^\sharp(D_+(x_i x_j)) : R[x_0, \dots, x_j = 1, \dots, x_n]_{x_i} \rightarrow R[x_0, \dots, x_i = 1, \dots, x_n]_{x_j}, \quad x_k \mapsto \frac{x_k}{x_j}.$$

**Definition 4.25.** Let  $\mathcal{C}$  be any category and  $f : X \rightarrow S$ ,  $g : Y \rightarrow S$  be two morphisms in  $\mathcal{C}$ . The fibred product  $X \times_S Y$  is an object in  $\mathcal{C}$  together with projection morphisms  $p_1 : X \times_S Y \rightarrow X$  and  $p_2 : X \times_S Y \rightarrow Y$  such that  $f \circ p_1 = g \circ p_2$  and for every object  $Z$  with morphisms  $\alpha : Z \rightarrow X$ ,

$\beta : Z \rightarrow Y$  such that  $f \circ \alpha = g \circ \beta$ , there exists a unique morphism  $\theta : Z \rightarrow X \times_S Y$  such that  $p_1 \circ \theta = \alpha$  and  $p_2 \circ \theta = \beta$ .



**Example 4.26.** The category of sets has all fibred products:  $X \times_S Y \cong \{(x, y) \in X \times Y \mid f(x) = g(y)\}$ . Some special situations: If  $|S| = 1$ , then  $f(x) = g(y)$  is automatically satisfied, so  $X \times_S Y = X \times Y$ . If  $f$  and  $g$  are inclusions, then  $X \times_S Y \cong X \cap Y$ . If  $f$  is arbitrary and  $g : \{s\} \hookrightarrow S$ , then  $X \times_S Y \cong f^{-1}(s)$ .

**Lemma 4.27.** If the fibred product  $X \times_S Y$  of  $f : X \rightarrow S$ ,  $g : Y \rightarrow S$  in  $\mathcal{C}$  exists, then it is unique up to unique isomorphism.

Lecture 19  
Jun 25, 2026

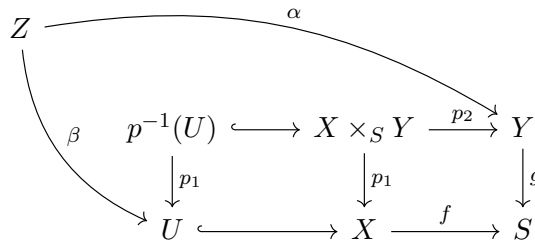
*Proof.* The usual abstract nonsense argument.  $\square$

**Lemma 4.28.** Let  $f : X \rightarrow Y$  be a morphism of schemes, and  $U \subseteq Y$  be open such that  $\text{im } f \subseteq U$ . Then there is a unique morphism  $g : X \rightarrow U$  such that  $f = i_U \circ g$ , where  $i_U : U \rightarrow Y$  is the open immersion.

*Proof.* Clear on the level of topological spaces. Set  $g_W^\# = f_W^\#$  for  $W \subseteq U$ . Then for  $V \subseteq Y$  open,  $(i_* g^\#)_V \circ i_V^\# = f_{V \cap U}^\# \circ i_V^\# = f_V^\#$  since  $\text{im } f \subseteq U$ .  $\square$

**Proposition 4.29.** Let  $f : X \rightarrow S$ ,  $g : Y \rightarrow S$  be morphisms of schemes. Assume that  $X \times_S Y$  exists. Let  $U \subseteq X$  be open. Then  $p_1^{-1}(U)$  is the fibred product  $U \times_S Y$ .

*Proof.* Consider the diagram



The rectangle clearly commutes, and for any  $Z$  as given, by assumption there is a unique morphism  $\theta : Z \rightarrow X \times_S Y$ . Note that  $\text{im}(p_1 \circ \theta) \subseteq U$ , hence  $\text{im } \theta \subseteq p^{-1}(U)$ , so by the previous lemma we get a unique morphism  $Z \rightarrow p^{-1}(U)$  as required (note that by uniqueness in the previous lemma, open immersions are monomorphisms).  $\square$

**Theorem 4.30.** The category of schemes has all fibred products.

*Proof.* Let  $f : X \rightarrow S$ ,  $g : Y \rightarrow S$ . First assume  $X = \text{Spec } A$ ,  $Y = \text{Spec } B$  and  $S = \text{Spec } R$  are all affine. Then by sheet 4, exercise 4,  $\text{Spec}(A \otimes_R B)$  (with the projection morphisms induced by  $A \rightarrow A \otimes_R B$ ,  $B \rightarrow A \otimes_R B$ ) is the fibred product  $X \times_S Y$ , since  $A \otimes_R B$  satisfies precisely the dual universal property in the category of commutative rings.

Now let  $X, Y, S$  be arbitrary. Let  $X = \bigcup_i U_i$  be an open cover and assume that all  $U_i \times_S Y$  exist. Then  $X \times_S Y$  exists as well: For every  $i, j$ , let  $p_i : U_i \times_S Y \rightarrow U_i$  be the first projection,

and set  $X_{ij} = p_i^{-1}(U_i \cap U_j)$ . Then by proposition 4.29,  $X_{ij}$  and  $X_{ji}$  both are fibred products  $(U_i \cap U_j) \times_S Y$ , so by lemma 4.27 there exists a unique isomorphism  $\varphi_{ij} : X_{ij} \rightarrow X_{ji}$ . For  $i, j, k$  write  $U_{ijk} = U_i \cap U_j \cap U_k$ . Then  $p_i^{-1}(U_{ijk}) = X_{ij} \cap X_{ik}$ , hence  $\varphi_{ij}(X_{ij} \cap X_{ik}) = p_j^{-1}(U_{ijk}) = X_{ji} \cap X_{jk}$ . Finally,  $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$  by lemma 4.27, since both sides are isomorphisms  $(U_{ijk}) \times_S Y \cong X_{ij} \cap X_{ik} \rightarrow X_{ki} \cap X_{kj} \cong (U_{ijk}) \times_S Y$ . Hence by proposition 4.22 the  $U_i \times_S Y$  glue along the  $\varphi_{ij}$  to a scheme  $Z = \bigcup_i V_i$  with compatible isomorphisms  $\psi_i : U_i \times_S Y \rightarrow V_i$ . It remains to check that  $Z$  is the fibred product of  $X$  and  $Y$  over  $S$ .

The projections  $V_i \cong U_i \times_S Y \rightarrow U_i \hookrightarrow X$  agree on their intersection by construction, so by proposition 2.25 they glue to a morphism  $p : Z \rightarrow X$ . Similarly one gets the projection  $q : Z \rightarrow Y$ . That  $f \circ p = g \circ q$  can be checked locally, where it holds. Now let  $\alpha : Q \rightarrow X$ ,  $\beta : Q \rightarrow Y$  be morphisms of schemes with  $f \circ \alpha = g \circ \beta$ . Let  $Q_i := \alpha^{-1}(U_i)$ , then we get unique morphisms  $\theta_i : Q_i \rightarrow U_i \times_S Y \hookrightarrow Z$  compatible with  $p_i$  and  $q_i$ . By uniqueness,  $\theta_i|_{U_i \cap U_j} = \theta_j|_{U_i \cap U_j}$ , so the  $\theta_i$  glue to a morphism  $\theta : Q \rightarrow Z$ . That  $\theta$  commutes with the projections and is unique can be checked locally, where it is clear. This proves  $Z = X \times_S Y$ .

Then if  $S$  is affine, applying the above argument to an affine open cover of  $X$  and, separately,  $Y$ , one finds that  $X \times_S Y$  exists. Finally, let  $S$  be an arbitrary scheme and take an affine open cover  $S = \bigcup_i S_i$ . Set  $X_i := f^{-1}(S_i)$ ,  $Y_i := g^{-1}(S_i)$ . Then  $X_i \times_{S_i} Y_i$  exists for all  $i$ . Notice that  $X_i \times_{S_i} Y_i$  is a fibred product  $X_i \times_S Y$ . Indeed, given  $\alpha : Z \rightarrow X_i$ ,  $\beta : Z \rightarrow Y$  with  $f \circ \alpha = g \circ \beta$ , we must have  $\text{im } \beta \subseteq Y_i$ , so by lemma 4.28  $\beta$  factors uniquely through  $\beta_i : Z \rightarrow Y_i$ , and the universal property applied to  $\alpha, \beta_i$  yields the desired unique map  $Z \rightarrow X_i \times_{S_i} Y_i$ . Therefore all the  $X_i \times_S Y$  exist, and one final application of the previous step shows that  $X \times_S Y$  exists.  $\square$

**Example 4.31.** Let  $R$  be a ring. Then  $\mathbb{A}_R^n \otimes_R \mathbb{A}_R^m = \text{Spec } R[x_1, \dots, x_n] \otimes_R R[y_1, \dots, y_m] \cong R[x_1, \dots, x_n, y_1, \dots, y_m] = \mathbb{A}_R^{n+m}$ . Note that here we wrote  $\otimes_R$  instead of  $\otimes_{\text{Spec } R}$  for convenience. More generally, if  $X$  and  $Y$  are affine schemes of finite type over a ring  $R$ , say  $X = \text{Spec } R[x_1, \dots, x_n]/(f_1, \dots, f_k)$  and  $Y = \text{Spec } R[y_1, \dots, y_m]/(g_1, \dots, g_l)$ , then  $X \times_R Y = \text{Spec } R[x_1, \dots, x_n, y_1, \dots, y_m]/(f_1, \dots, f_k, g_1, \dots, g_l)$ .

**Remark 4.32.** Note that if  $X, Y \rightarrow S$  are morphism of schemes, we have a map of sets  $|X \times_S Y| \rightarrow |X| \otimes_{|S|} |Y|$ , where  $|Z|$  is the underlying space of  $Z$ . But this map is not a bijection in general. For example, if  $k$  is algebraically closed, then it is easy to see that  $|\mathbb{A}_k^2| \rightarrow |\mathbb{A}_k^1| \times |\mathbb{A}_k^1|$  is not an isomorphism. Or if  $X = Y = \text{Spec } \mathbb{C}$  and  $S = \text{Spec } \mathbb{R}$ , then  $|X| \times |Y|$  is a point, but  $X \times_S Y \cong \text{Spec } \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \text{Spec } \mathbb{C}[X]/(X^2 + 1) \cong \text{Spec } \mathbb{C} \times \mathbb{C} \cong \text{Spec } \mathbb{C} \sqcup \text{Spec } \mathbb{C}$  has two points.

**Definition 4.33.** Let  $f : X \rightarrow Y$  be a morphism of schemes and  $y \in Y$ . Let  $\text{Spec } k(y) \rightarrow Y$  be the morphism associated to  $y$ . The fibred product  $X_y := X \times_Y \text{Spec } k(y)$  is called the *fibre of  $f$  over  $y$* .

**Proposition 4.34.** As topological spaces,  $X_y$  is homeomorphic to  $f^{-1}(\{y\}) \subseteq X$ .

*Proof.* Exercise.  $\square$

**Example 4.35.** In example 4.26 we saw that in the category of sets, the fibre product of two subsets is their intersection. If  $X_1, X_2 \subseteq X$  are open subschemes, then in the same way  $X_1 \times_X X_2 = X_1 \cap X_2$ , since by lemma 4.28 the intersection satisfies the universal property of the fibred product.

Now let  $i : X_1 \hookrightarrow X, j : X_2 \hookrightarrow X$  be closed subschemes of a scheme  $X$ . We define their scheme-theoretic intersection as  $X_1 \times_X X_2$ . Write  $p, q$  for the projections and let  $\pi = i \circ p = j \circ q$ . We show that  $\pi$  is a closed immersion again. This can be done locally, so let  $\text{Spec } A \cong U \subseteq X$  be an affine open. By proposition 4.9 we have  $i^{-1}(U) \cong \text{Spec } A/I$  and  $j^{-1}(U) \cong \text{Spec } A/J$  for some ideals  $I, J \subseteq A$ . Hence  $\pi^{-1}(U) \cong i^{-1}(U) \times_U j^{-1}(U) \cong \text{Spec } A/I \otimes_A A/J \cong \text{Spec } A/(I+J)$ , so by corollary 4.10  $\pi$  is a closed immersion, and since  $V(I) \cap V(J) = V(I+J)$  we see that on topological spaces, this indeed gives the intersection, cf. remark 4.18.

**Proposition 4.36.** *Let  $S$  be a scheme and  $X, Y, Z$  schemes over  $S$ . Then*

- (i) *There are canonical isomorphisms of schemes  $X \times_S S \rightarrow X$ ,  $X \times_S Y \rightarrow Y \times_S X$  and  $(X \times_S Y) \times_S Z \cong X \times_S (Y \times_S Z) =: X \times_S Y \times_S Z$ .*
- (ii)  *$X \times_Y (Y \times_S Z) \cong X \times_S Z$ .*
- (iii) *Let  $f : X \rightarrow X'$ ,  $g : Y \rightarrow Y'$  be morphisms of  $S$ -schemes. Then there exists a unique morphism  $f \times g : X \times_S Y \rightarrow X' \times_S Y'$  such that the following diagram commutes:*

$$\begin{array}{ccccc}
 & & Y & \xrightarrow{g} & Y' \\
 & \nearrow & & & \nearrow \\
 X \times_S Y & \xrightarrow{f \times g} & X' \times_S Y' & & \\
 & \searrow & & & \searrow \\
 & & X & \xrightarrow{f} & X' \\
 & & & & \nearrow \\
 & & & & S
 \end{array}$$

*Proof.* Everything follows immediately from the universal property.  $\square$

**Base change** Let  $S$  be a scheme and  $X, S'$  be schemes over  $S$ . Then  $X_{S'} := X \times_S S'$  can be considered an  $S'$ -scheme using the second projection. This process is called base change of  $X$  by  $S' \rightarrow S$ . If  $f : X \rightarrow Y$  is a morphism of  $S$ -schemes, then  $f_{S'} := f \times \text{id}_{S'} : X_{S'} \rightarrow Y_{S'}$  is an induced morphism of the base-changed schemes, making base change a functor.

Once again abusing notation, if  $S' = \text{Spec } B$  is affine, we write  $X_B := X_{\text{Spec } B}$  and  $f_B := f_{\text{Spec } B}$ .

**Example 4.37.** Let  $S = \text{Spec } A$  and  $X = \text{Spec } A[x_1, \dots, x_n]/(f_1, \dots, f_m)$  an affine scheme of finite type over  $A$ . If  $B$  is an  $A$ -algebra, then we can base change along  $S' := \text{Spec } B \rightarrow \text{Spec } A$  and obtain  $X_B = \text{Spec } A[x_1, \dots, x_n]/(f_1, \dots, f_m) \otimes_A B \cong \text{Spec } B[x_1, \dots, x_n]/(f_1, \dots, f_m)$ .

**Definition 4.38.** A property  $\mathcal{P}$  of a morphism of schemes is *stable under base change* if for every morphism  $X \rightarrow Y$  with  $\mathcal{P}$  and for every scheme  $Y'$  over  $Y$ , the morphism  $X \times_Y Y' \rightarrow Y'$  also satisfies  $\mathcal{P}$ .

**Proposition 4.39.** *Open and closed immersions are stable under base change.*

*Proof.* Let  $X \rightarrow Y$  be an open immersion. By proposition 4.29,  $X \times_Y Y' \cong p^{-1}(X)$  is an open subscheme of  $Y \times_Y Y' \cong Y'$ .

Now let  $i : X \hookrightarrow Y$  be a closed immersion, and  $f : Y' \rightarrow Y$  any morphism. Let  $\text{Spec } A \cong S \subseteq Y$  be an affine open. Then by proposition 4.9,  $i^{-1}(S) \cong \text{Spec } A/I$  for some ideal  $I \subseteq A$ . Let  $\text{Spec } B \cong V \subseteq f^{-1}(S)$  be some affine open. Then by construction of the fibred product,  $X \times_Y Y'$  is covered by affine opens of the form  $i^{-1}(S) \times_S V \cong \text{Spec } A/I \otimes_A B \cong \text{Spec } B/IB$ , so by corollary 4.10,  $X \times_Y Y' \rightarrow Y'$  is a closed immersion.  $\square$

**Remark 4.40.** Let  $i : X \rightarrow Y$  be a closed immersion and  $f : Y' \rightarrow Y$  a morphism of schemes. The above proof shows that  $\text{im}(X \times_Y Y' \rightarrow Y) = f^{-1}(X)$ , so  $X \times_Y Y' \cong f^{-1}(X)$  as closed subsets. In particular, if  $y$  is a closed point in  $Y$ , then this matches up with definition 4.33.

## 5 Separated and Proper Morphisms

**Remark 5.1.** Per definition, a topological space  $X$  is Hausdorff if any two points can be separated by open neighbourhoods. This is equivalent to the diagonal  $\Delta_X = \{(x, y) \in X \times X \mid x = y\}$  being a closed subset of  $X \times X$ .

**Remark 5.2.** Let  $X$  be a scheme. Its underlying topological space is almost never Hausdorff. For example, if  $X$  is a variety, then  $X$  is Hausdorff if and only if it is finite.

**Definition 5.3.** A morphism  $f : X \rightarrow Y$  of schemes is *separated* if the diagonal morphism  $\Delta : X \rightarrow X \times_Y X$  is a closed immersion. If  $A$  is a ring, then an  $A$ -scheme  $X$  is separated (over  $A$ ) if the structure morphism  $X \rightarrow \operatorname{Spec} A$  is separated.

**Remark 5.4.** To check that  $\Delta$  is a closed immersion, it is enough to check that  $\operatorname{im} \Delta$  is closed in  $X \times X$  (Exercise). While this definition was motivated by remark 5.1, a separated  $k$ -scheme  $X$  is usually not Hausdorff since in general  $|X \times_k X| \not\cong |X| \times |X|$ , see remark 4.32.

**Remark 5.5.** All “reasonable” schemes are separated. Non-separated schemes can be considered pathological examples.

**Proposition 5.6.** *Any morphism of affine schemes is separated.*

*Proof.* Let  $f : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$  be a morphism of affine schemes. Then  $\Delta$  is induced by the ring morphism  $B \otimes_A B \rightarrow B, b_1 \otimes b_2 \mapsto b_1 b_2$ , which is clearly surjective. By remark 4.8,  $\Delta$  is a closed immersion.  $\square$

**Proposition 5.7.** *Let  $A$  be a ring.  $\mathbb{P}_A^n$  is a separated  $A$ -scheme.*

*Proof.* Exercise.  $\square$

**Proposition 5.8.** (i) *Any open or closed immersion is separated.*

(ii) *A composition of separated morphisms is separated.*

(iii) *Separated morphisms are stable under base change.*

(iv) *Let  $f : X \rightarrow Y, g : Y \rightarrow Z$  be morphisms of schemes such that  $g \circ f$  is separated. Then  $f$  is separated.*

(v) *If  $f : X \rightarrow Y$  is separated, then so is  $f|_{f^{-1}(V)} : f^{-1}(V) \rightarrow V$  for any  $V \subseteq Y$  open.*

**Remark 5.9.** From the above propositions we deduce immediately that all morphisms between open or closed subschemes of projective spaces are separated.

**Example 5.10** (Affine line with double point). Let  $k$  be a field and glue two copies of  $\mathbb{A}_k^1$  along  $\operatorname{id} : \mathbb{A}_k^1 \setminus \{0\} \rightarrow \mathbb{A}_k^1 \setminus \{0\}$ . The resulting scheme  $X$  is not separated over  $k$  (Exercise).

**Proposition 5.11.** *Assume  $f : X \rightarrow \operatorname{Spec} A$  is a separated morphism to an affine scheme and  $U, V \subseteq X$  are two open affine subschemes. Then  $U \cap V$  is also affine.*

*Proof.* We have  $U \cap V = \Delta^{-1}(U \times_A V)$ , where  $U \times_A V$  is affine by construction. So the claim follows from proposition 4.9.  $\square$

**Definition 5.12.** A *variety* over a field  $k$  is a reduced, separated scheme of finite type over  $k$ .

**Definition 5.13.** (i) A morphism  $(f, f^\#) : X \rightarrow Y$  of schemes is *closed* if the morphism of underlying spaces  $f : X \rightarrow Y$  is.

(ii) A morphism  $X \rightarrow Y$  of schemes is *universally closed* if for every  $Z \rightarrow Y$ , the base change morphism  $Z \times_Y X \rightarrow Z$  is closed.



**Definition 5.14.** A morphism of schemes  $f : X \rightarrow Y$  is *proper* if it is separated, of finite type, and universally closed.

**Remark 5.15.** In definition 3.22 we only defined being of finite type for morphisms  $X \rightarrow \operatorname{Spec} A$ . More generally, a morphism  $f : X \rightarrow Y$  is of finite type if there exists an open affine cover  $Y = \bigcup_i \operatorname{Spec} A_i$  such that  $f^{-1}(\operatorname{Spec} A_i) \rightarrow \operatorname{Spec} A_i$  is of finite type.

**Remark 5.16.** In topology, a morphism  $f : X \rightarrow Y$  is proper if the preimage of every compact set is compact. In particular,  $X$  is compact if and only if  $X \rightarrow \{*\}$  is proper. For nice topological spaces, being proper is equivalent to being universally closed. Hence “properness” is a suitable replacement for compactness for schemes.

**Example 5.17.** Closed immersions are proper by propositions 5.8 and ??.

**Example 5.18.** Let  $k$  be algebraically closed. Then  $\mathbb{A}_k^1 \rightarrow \operatorname{Spec} k$  is not proper over  $k$  because it is not universally closed:  $\mathbb{A}_k^2 \rightarrow \mathbb{A}_k^1$  sends  $V(xy - 1)$  to  $\mathbb{A}^1 \setminus \{0\}$ .

**Proposition 5.19.** (i) *Compositions of proper morphisms are proper*  
(ii) *Proper morphisms are stable under base change*  
(iii)  *$f : X \rightarrow Y$  is proper if and only for every affine open cover  $Y = \bigcup_i U_i$ ,  $f : f^{-1}(U_i) \rightarrow U_i$  is proper for all  $i$ .*

**Definition 5.20.** (i) Let  $A$  be ring. We defined  $\mathbb{P}_A^n = \operatorname{Proj} A[x_0, \dots, x_n] = \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} \operatorname{Spec} A$ . Similarly, for a scheme  $X$ , we can define  $\mathbb{P}_X^n := \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} X$  to be projective  $n$ -space over  $X$ .  
(ii) A morphism  $f : X \rightarrow Y$  is projective if it factors as  $f = p \circ i$  with  $i : X \rightarrow \mathbb{P}_Y^n$  a closed immersion and  $p : \mathbb{P}_Y^n \rightarrow Y$  the canonical projection. As usual, an  $A$ -scheme  $X$  is called projective if the structure morphism  $X \rightarrow \operatorname{Spec} A$  is projective, that is if  $X$  is a closed subscheme of  $\mathbb{P}_A^n$ .

**Theorem 5.21.** *Any projective morphism  $f : X \rightarrow Y$  is proper.*

*Proof (Sketch).* By example 5.17 and proposition 5.19 it suffices to show that  $\mathbb{P}_{\mathbb{Z}}^n \rightarrow \operatorname{Spec} \mathbb{Z}$  is universally closed. For this the same proof as in classical algebraic geometry works (see last semester).  $\square$

## Dimension

**Definition 5.22.** The *dimension*  $\dim X$  of a scheme is the dimension of its underlying topological space.

**Proposition 5.23.** *For a ring  $A$ ,  $\dim \operatorname{Spec} A$  equals the Krull dimension of  $A$ .*

**Example 5.24.**  $\dim \mathbb{A}_R^n = n + \dim R$ ,  $\dim \mathbb{Z} = 1$  and  $\dim k[x]/(x^2) = 0$ .

**Proposition 5.25.** *Let  $X$  be an integral scheme of finite type over a field  $k$ .*

- (i) *For every non-empty open  $U \subseteq X$ ,  $\dim X = \dim U$ .*
- (ii) *Let  $U = \operatorname{Spec} A \subseteq X$  be affine. Then  $\dim U = \dim A = \operatorname{tr. deg}_k \operatorname{Frac} A$ .*

## 6 $\mathcal{O}_X$ -Modules and Coherent Sheaves

**Definition 6.1.** Let  $(X, \mathcal{O}_X)$  be a ringed space.

- (i) A sheaf of  $\mathcal{O}_X$ -modules is a sheaf of groups  $\mathcal{F}$  on  $X$  such that for every  $U \subseteq X$  open,  $\mathcal{F}(U)$  is a  $\mathcal{O}_X(U)$ -module and this structure is compatible with restrictions, that is, for  $a \in \mathcal{O}_X(U)$  and  $m \in \mathcal{F}(U)$ , one has  $(a \cdot m)|_V = a|_V \cdot m|_V$  for all  $V \subseteq U$  open.
- (ii) A morphism  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  of  $\mathcal{O}_X$ -modules is a morphism of sheaves such that  $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$  is a morphism of  $\mathcal{O}_X(U)$ -modules for all  $U \subseteq X$  open.

**Remark 6.2.** If  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  is a morphism of  $\mathcal{O}_X$ -modules, then  $\ker \varphi$ ,  $\operatorname{im} \varphi$  and  $\operatorname{coker} \varphi$  are  $\mathcal{O}_X$ -modules. Subsheaves of  $\mathcal{O}_X$ -modules are  $\mathcal{O}_X$ -modules. If  $\mathcal{F}, \mathcal{G}$  are  $\mathcal{O}_X$ -modules, so is  $\mathcal{F} \oplus \mathcal{G}$ . Finally, define  $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$  as the sheafification of  $U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$ .

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**Definition 6.3.** Let  $f : X \rightarrow Y$  be a morphism of ringed spaces, and let  $\mathcal{F}$  be an  $\mathcal{O}_X$ -module. Then  $f_*\mathcal{F}$  is naturally a  $f_*\mathcal{O}_X$ -module, hence also an  $\mathcal{O}_Y$ -module via restriction of scalars.

**Definition 6.4.** An  $\mathcal{O}_X$ -module  $\mathcal{F}$  is called *free* if  $\mathcal{F} \cong \bigoplus_I \mathcal{O}_X$ .  $\mathcal{F}$  is called *locally free* if there is an open cover  $X = \bigcup_i U_i$  such that  $\mathcal{F}|_{U_i}$  is a free  $\mathcal{O}_{U_i}$ -module for every  $i$ .

**Definition 6.5.** Let  $A$  be a ring and  $M$  be an  $A$ -module. We define a Spec  $A$ -module  $\widetilde{M}$  via

$$\widetilde{M}(U) := \left\{ s : U \rightarrow \prod_{\mathfrak{p} \in U} M_{\mathfrak{p}} \mid s(\mathfrak{p}) \in M_{\mathfrak{p}} \text{ for all } \mathfrak{p} \in U, \text{ and } (*) \right\}$$

for all  $U \subseteq X$  open, where  $(*)$  is (as usual) the following condition: For every  $\mathfrak{p} \in U$  there exists an open neighbourhood  $V \subseteq U$  such that  $s(\mathfrak{q}) = \frac{m}{f} \in M_{\mathfrak{q}}$  for all  $\mathfrak{q} \in V$ , where  $m \in M$  and  $f \in A$ ,  $f \notin \mathfrak{q}$  for all  $\mathfrak{q} \in V$ .

Note that this is essentially the same definition as for the structure sheaf of an affine scheme, just with  $A$  replaced by  $M$ . Hence the exact same proofs show

**Proposition 6.6.** Let  $A$  be a ring and  $M$  an  $A$ -module.

- (i)  $\widetilde{M}_{\mathfrak{p}} = M_{\mathfrak{p}}$  for all  $\mathfrak{p} \in \operatorname{Spec} A$ .
- (ii)  $\widetilde{M}(D(f)) = M_f$  for all  $f \in A$ .

**Definition 6.7.** A sequence of sheaves  $\mathcal{F} \xrightarrow{\varphi} \mathcal{G} \xrightarrow{\psi} \mathcal{H}$  is *exact* at  $\mathcal{G}$  if  $\ker \psi = \operatorname{im} \varphi$ . Equivalently, the sequence is exact if and only if the induced sequence  $\mathcal{F}_x \rightarrow \mathcal{G}_x \rightarrow \mathcal{H}_x$  of stalks is exact for all  $x \in X$ . (Recall the sheafification in the definition of image, so this is not equivalent to the induced sequence on every open  $U \subseteq X$  being exact.)

**Proposition 6.8.** The functor  $M \mapsto \widetilde{M}$  from  $A$ -modules to Spec  $A$ -modules is exact and fully faithful.

*Proof.* If  $\varphi : M \rightarrow N$  is an  $A$ -module morphism, we set  $\widetilde{\varphi} : \widetilde{M} \rightarrow \widetilde{N}$  as  $\widetilde{\varphi}_U(s)(\mathfrak{p}) = \varphi_{\mathfrak{p}}(s(\mathfrak{p}))$ . Exactness follows from the exactness of localization, faithfulness is clear since  $\widetilde{\varphi}_X = \varphi$ , and for fullness we have to check that a module morphism  $\widetilde{M} \rightarrow \widetilde{N}$  is uniquely specified by the map of global sections. By compatibility with the module structure and proposition 6.6, this is clear on stalks (or on a basis), hence the claim follows.  $\square$

**Remark 6.9.** For  $A$ -modules  $M$  and  $N$  one has  $\widetilde{M \oplus N} = \widetilde{M} \oplus \widetilde{N}$  and  $\widetilde{M \otimes_A N} \cong \widetilde{M} \otimes_{\mathcal{O}_{\operatorname{Spec} A}} \widetilde{N}$ .

**Definition 6.10.** An  $\mathcal{O}_X$ -module  $\mathcal{F}$  is *quasi-coherent* if there exists an open affine cover  $X = \bigcup_i U_i$ ,  $U_i \cong \operatorname{Spec} A_i$  such that  $\mathcal{F}|_{U_i} \cong \widetilde{M_i}$  for some  $A_i$ -module  $M_i$ .

**Proposition 6.11.** An  $\mathcal{O}_X$ -module is quasi-coherent if and only if for every open affine  $\operatorname{Spec} A \cong U \subseteq X$  one has  $\mathcal{F}|_U \cong \widetilde{M}$  for some  $A$ -module  $M$ .

*Proof (Idea).* Verify the conditions of the Affine Communication Lemma 3.19.  $\square$

**Corollary 6.12.** The functor from proposition 6.8 is an equivalence of categories between  $A$ -modules and quasi-coherent  $\operatorname{Spec} A$ -modules.

*Proof.* By the proposition it is essentially surjective, and we already knew that it is fully faithful.  $\square$

**Quasi-Coherent Modules on  $\operatorname{Proj} B$ .** Let  $B$  be a graded ring and  $M$  a graded  $B$ -module, that is  $M = \bigoplus_{n \in \mathbb{Z}} M_n$  is a graded abelian group with a  $B$ -module structure, and  $B_n \cdot M_m \subseteq M_{n+m}$ .

**Definition 6.13.** We define a  $\operatorname{Proj} B$ -module  $\widetilde{M}$  via

$$\widetilde{M}(U) = \left\{ s : U \rightarrow \prod_{\mathfrak{p} \in U} M_{(\mathfrak{p})} \mid s(\mathfrak{p}) \in M_{(\mathfrak{p})} \text{ for all } \mathfrak{p} \in U, \text{ and } (*) \right\}$$

for all  $U \subseteq X$  open, where  $(*)$  is (as usual) the following condition: For every  $\mathfrak{p} \in U$  there exists an open neighbourhood  $V \in \mathcal{B}_U$  such that  $s(\mathfrak{q}) = \frac{m}{f} \in M_{(\mathfrak{q})}$  for all  $\mathfrak{q} \in V$ , where  $m \in M$  and  $f \in A$ ,  $f \notin \mathfrak{q}$  for all  $\mathfrak{q} \in V$ , and  $m$  and  $f$  are homogeneous of the same degree.

**Proposition 6.14.** Let  $B$  be a graded ring and  $M$  a graded  $B$ -module.

- (i)  $\widetilde{M}_{\mathfrak{p}} = M_{(\mathfrak{p})}$  for every  $\mathfrak{p} \in \operatorname{Proj} B$ .
- (ii) For every  $f \in B_+$  there is an isomorphism  $\widetilde{M}|_{D_+(f)} \cong \widetilde{M_{(f)}}$ , where  $M_{(f)}$  is taken to be the (ungraded)  $B_{(f)}$ -module in the natural way.
- (iii)  $\widetilde{M}$  is quasi-coherent.

*Proof.* (i) and (ii): Replace  $B$  with  $M$  in the respective proofs for  $\operatorname{Proj} B$ .

(iii) follows directly from (ii).  $\square$

**Example 6.15.** Let  $B$  be a graded ring and denote by  $B(n)$  the  $B$ -module  $B$  with shifted grading  $B(n)_d = B_{n+d}$ . Write  $\mathcal{O}_{\operatorname{Proj} B}(n) := \widetilde{B(n)}$  for the associated quasi-coherent  $\operatorname{Proj} B$ -module. For example, if  $X = \mathbb{P}_A^n$ , then  $\mathcal{O}_X(m)$  has as global sections the homogeneous polynomials of degree  $m$  in  $A[x_0, \dots, x_n]$ .